2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

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Abstract

This report is generated directly from the submitted CSV, JSON, and VTU files. It describes the governing equations and nondimensionalization, the cell-centred finite-volume discretization and boundary conditions, the implicit and transient algorithms reported in metadata, the MPI partition and halo strategy, the output and sanity-check workflow, and the per-case results. The analysis uses the full required-case scope under `results`. Of the 8 required cases, 6 are assessed as **converged**; 2 are assessed as **statistically periodic**. Every selected case passes the automated physics and figure checks. Solver metadata identify: cfd2d.

1 Introduction

The benchmark asks for a cell-centred, unstructured, compressible Navier–Stokes solver with second-order reconstruction, a limiter, an approximate Riemann flux, implicit pseudo-time marching, a true physical-time transient path for the cylinder at Reynolds number 200, METIS partitioning, and neighbour-scoped halo exchange. This document is the technical report for the submitted result packages. It is generated from the actual artifacts rather than from a hand-written results table, so missing data remain visible as failures.

The read-only specification is in `cfd_solver_agentic_benchmark/`; the post-processing tools are in `solver/tools/`, and the result directories are under `results`. The selected cases are:

1. `naca0012_m015_inviscid` (mesh `NACA0012_H2.cgns`; declared **statistically periodic**).
2. `naca0012_m080_inviscid` (mesh `NACA0012_H2.cgns`; declared **converged**).
3. `naca0012_m200_inviscid` (mesh `NACA0012_H2.cgns`; declared **converged**).
4. `naca0012_m015_laminar_re5000` (mesh `NACA0012_H2.cgns`; declared **converged**).
5. `naca0012_m080_laminar_re5000` (mesh `NACA0012_H2.cgns`; declared **converged**).
6. `naca0012_m200_laminar_re5000` (mesh `NACA0012_H2.cgns`; declared **converged**).
7. `cylinder_m010_laminar_re20` (mesh `CylinderB1.cgns`; declared **converged**).
8. `cylinder_m010_laminar_re200` (mesh `CylinderB1.cgns`; declared **statistically periodic**).

2 Governing Equations and Nondimensionalization

The conservative state is

$$\mathbf{U} = [\rho,\ \rho u,\ \rho v,\ \rho E]^T, \tag{1}$$

and the two-dimensional compressible Navier–Stokes equations are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}. \tag{2}$$

The inviscid fluxes are

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho u v \\ u(\rho E + p) \end{bmatrix}, \qquad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho u v \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}. \tag{3}$$

The calorically perfect gas closure is

$$p = (\gamma - 1)\rho e, \qquad E = e + \tfrac{1}{2}\left(u^2 + v^2\right), \qquad a = \sqrt{\gamma p/\rho}, \tag{4}$$

with the gas constants supplied in the case files. For viscous cases the Newtonian stress tensor and Fourier heat flux are

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu\left(u_x + v_y\right), \qquad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu\left(u_x + v_y\right), \qquad \tau_{xy} = \mu\left(u_y + v_x\right), \tag{5}$$

$$\mathbf{q} = -k\nabla T, \qquad k = \frac{\mu c_p}{Pr}, \qquad c_p = \frac{\gamma R}{\gamma - 1}. \tag{6}$$

For the constant-viscosity laminar runs,

$$\mu = \frac{\rho_\infty U_\infty L_{\mathrm{ref}}}{Re}, \tag{7}$$

so the supplied Reynolds number and reference length fix the viscosity.

The nondimensional reference scales are length L_{ref}, velocity U_∞, density ρ_∞, and pressure

$$p_\infty = \frac{\rho_\infty U_\infty^2}{\gamma M_\infty^2}. \tag{8}$$

The dynamic pressure is $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2$. Lengths are scaled by L_{ref}, velocities by U_∞, time by $L_{\mathrm{ref}}/U_\infty$, and pressure by $\rho_\infty U_\infty^2$. Force and moment coefficients are

$$C_D = \frac{D}{q_\infty A_{\mathrm{ref}}}, \qquad C_L = \frac{L}{q_\infty A_{\mathrm{ref}}}, \qquad C_m = \frac{M}{q_\infty A_{\mathrm{ref}} L_{\mathrm{ref}}}, \tag{9}$$

and the surface pressure coefficient is

$$C_p = \frac{p - p_\infty}{q_\infty}. \tag{10}$$

The reported force columns separate pressure and viscous contributions. The skin-friction coefficient uses the tangential wall shear, $c_f = \tau_w/q_\infty$, rather than the full viscous normal traction.

3 Meshes, Boundary Tags, and Case Inputs

The meshes are supplied as CGNS unstructured zones. Boundary families are read from the mesh metadata and mapped through each case JSON to farfield, inviscid slip wall, or viscous no-slip adiabatic wall conditions. Cell volumes are computed from the closed polygon faces; face areas and normals are computed from the face vertices with a consistent outward orientation. Wall length integrals used by the force coefficients are accumulated over the wall faces. The case inputs and mesh sizes read from the result metadata are summarized in Table 1.

Table 1: Case inputs, boundary-family mapping, and global mesh size read from the case JSON and result metadata. A missing value means the artifact did not record it.

Case	Mesh	Ncell	M_∞	AoA	Re	Boundary tags (→ solver BC)	Mode / run type
`naca0012_m015 inviscid`	`NACA0012_ H2.cgns`	20816	0.15	0	—	`bc-2` → `farfield` `bc-4` → `slip_wall`	inviscid, steady
`naca0012_m080 inviscid`	`NACA0012_ H2.cgns`	20816	0.8	0	—	`bc-2` → `farfield` `bc-4` → `slip_wall`	inviscid, steady
`naca0012_m200 inviscid`	`NACA0012_ H2.cgns`	20816	2	0	—	`bc-2` → `farfield` `bc-4` → `slip_wall`	inviscid, steady
`naca0012_m015 laminar_re5000`	`NACA0012_ H2.cgns`	20816	0.15	0	5000	`bc-2` → `farfield` `bc-4` → `no_slip_ adiabatic_wall`	laminar, steady
`naca0012_m080 laminar_re5000`	`NACA0012_ H2.cgns`	20816	0.8	0	5000	`bc-2` → `farfield` `bc-4` → `no_slip_ adiabatic_wall`	laminar, steady
`naca0012_m200 laminar_re5000`	`NACA0012_ H2.cgns`	20816	2	0	5000	`bc-2` → `farfield` `bc-4` → `no_slip_ adiabatic_wall`	laminar, steady
`cylinder_m010 laminar_re20`	`CylinderB1. cgns`	10185	0.1	0	20	`WALL` → `no_slip_ adiabatic_wall` `FAR` → `farfield`	laminar, steady
`cylinder_m010 laminar_re200`	`CylinderB1. cgns`	10185	0.1	0	200	`WALL` → `no_slip_ adiabatic_wall` `FAR` → `farfield`	laminar, transient

The surface CSV is required to contain boundary-state values. For no-slip walls the reported velocity components should be near zero; for inviscid slip walls the normal component should be near zero while the tangential component may remain finite. The post-processing sanity gate checks those conditions from the submitted surface rows rather than substituting adjacent cell-centre values.

4 Spatial Discretization

The residual on cell i with volume V_i is assembled from face fluxes:

$$\frac{d\mathbf{U}_i}{dt} = -\frac{1}{V_i}\sum_{f\in\partial\Omega_i} A_f \hat{\mathbf{F}}_f \cdot \mathbf{n}_f + \frac{1}{V_i}\sum_{f\in\partial\Omega_i} A_f \hat{\mathbf{F}}^v_f \cdot \mathbf{n}_f, \tag{11}$$

where A_f is face area and $\mathbf{n}_f$ is the outward unit normal. The reported production metadata identify the reconstruction as weighted_least_squares_linear_primitive, the limiter as Barth_Jespersen, Barth_Jespersen_with_tolerance, and the claimed spatial order as 2.

4.1 Second-Order Reconstruction

For a primitive or conservative variable ϕ, the piecewise-linear reconstruction at face f is

$$\phi_f^- = \phi_i + \psi_i \nabla\phi_i \cdot (\mathbf{x}_f - \mathbf{x}_i)\,, \tag{12}$$

with the gradient obtained by a weighted least-squares solve

$$\mathbf{M}_i \nabla\phi_i = \sum_{j\in\mathcal{N}(i)} w_{ij}\,(\phi_j - \phi_i)\,(\mathbf{x}_j - \mathbf{x}_i)\,, \qquad \mathbf{M}_i = \sum_{j\in\mathcal{N}(i)} w_{ij}\,(\mathbf{x}_j - \mathbf{x}_i)\,(\mathbf{x}_j - \mathbf{x}_i)^T\,. \tag{13}$$

The same reconstruction is applied on the two sides of an interior face; the left and right states are then passed to the approximate Riemann solver. Where the metadata record a first-order fallback, that fallback and its trigger are identified in the per-case metadata rather than being hidden in the figures.

4.2 Limiter and Positivity Control

The metadata report the limiter as Barth_Jespersen, Barth_Jespersen_with_tolerance. The Barth–Jespersen reference form is shown here because it is the benchmark's minimum acceptable limiter; it is not attributed to a different limiter if the metadata name another method. For each cell the unlimited reconstructed values are bounded by the local neighbour extrema and the limiter factor is

$$\psi_i = \min_f \begin{cases} 1, & \phi_f^- = \phi_i, \\ \dfrac{\phi_i^{\max} - \phi_i}{\phi_f^- - \phi_i}, & \phi_f^- > \phi_i, \\ \dfrac{\phi_i^{\min} - \phi_i}{\phi_f^- - \phi_i}, & \phi_f^- < \phi_i, \end{cases} \tag{14}$$

clipped to $[0, 1]$ and applied to all components. The submitted positivity control is recorded as Barth_Jespersen_plus_first_order_face_fallback_plus_update_line_search. Density and pressure floors on reconstructed states and updates are the minimum acceptable safeguard; the sanity gate separately verifies that every final cell has positive density and pressure.

4.3 Inviscid and Viscous Fluxes

The metadata report the inviscid flux as rusanov, roe and the entropy treatment as harten. For the reported Roe flux the interface flux is assembled from the Euler flux difference projected on the Roe-averaged characteristic fields,

$$\hat{\mathbf{F}}_f \cdot \mathbf{n}_f = \frac{1}{2}\,(\mathbf{F}_L + \mathbf{F}_R) \cdot \mathbf{n}_f - \frac{1}{2}\sum_{k=1}^{4} \left|\tilde{\lambda}_k\right| \alpha_k \tilde{\mathbf{r}}_k, \tag{15}$$

where the tilde quantities use Roe averages of velocity and enthalpy. The metadata name a Harten-type entropy fix; in that form the acoustic eigenvalue used in $\left|\tilde{\lambda}_k\right|$ is replaced by

$$|\lambda|_\epsilon = \begin{cases} (\lambda^2 + \epsilon^2)/(2\epsilon), & |\lambda| < \epsilon, \\ |\lambda|, & |\lambda| \geq \epsilon, \end{cases} \tag{16}$$

with the entropy-fix parameter applied to the acoustic waves.

For a Rusanov/local Lax–Friedrichs interface flux the implemented form is

$$\hat{\mathbf{F}}_f \cdot \mathbf{n}_f = \frac{1}{2}\left(\mathbf{F}_L + \mathbf{F}_R\right) \cdot \mathbf{n}_f - \frac{1}{2}\alpha_f \left(\mathbf{U}_R - \mathbf{U}_L\right), \qquad \alpha_f = |\mathbf{u}_f \cdot \mathbf{n}_f| + a_f. \tag{17}$$

Viscous cases use inviscid: cell_gradient_face_corrected; laminar: cell_gradient_face_corrected. Primitive-variable gradients for velocity and temperature are reconstructed on the same unstructured stencil. The stress tensor and heat flux above are evaluated at faces from the averaged gradients, and wall-normal gradients are treated by the wall boundary condition. The pressure force is integrated as $p\mathbf{n}$ and kept separate from the tangential viscous force used for skin friction, so the force table's pressure and viscous drag columns are independent diagnostics.

5 Boundary Conditions

Farfield boundaries use a characteristic or Riemann treatment that preserves the supplied freestream state and allows outgoing waves to leave the domain. Inviscid slip walls enforce zero normal velocity while leaving the tangential velocity free. Viscous no-slip adiabatic walls enforce zero wall velocity and zero normal temperature gradient, with the wall shear evaluated tangentially. Surface output uses boundary values where the solver reports them; if a run instead reports adjacent cell-centre values, its metadata must say so and the report must not interpret those values as wall-condition values. The post-processing checks are:

- laminar surface rows: near-zero wall speed and nonzero skin friction;
- inviscid surface rows: near-zero normal velocity and negligible viscous force columns;
- every case: a nontrivial wall C_p variation.

6 Implicit and Transient Time Integration

Steady cases use pseudo-time marching with a local time step estimated from the convective and viscous spectral radii,

$$\Delta t_i = \mathrm{CFL}\,\frac{V_i}{\sum_{f\in\partial\Omega_i} A_f\left(|\mathbf{u}_f \cdot \mathbf{n}_f| + a_f\right) + \Delta t_i^v}, \tag{18}$$

where the viscous contribution is based on the face diffusivity. The nonlinear update is solved with the reported implicit method scalar_LU_SGS_rusanov_spectral_radius, scalar_LU_SGS_rusanov_spectral_radius_1
For LU-SGS the simplified Jacobian system is solved by a forward sweep and a backward sweep,

$$\left(\mathbf{D} + \mathbf{L}\right)\mathbf{D}^{-1}\left(\mathbf{D} + \mathbf{U}\right)\Delta\mathbf{U} = -\mathbf{R}, \tag{19}$$

with local spectral-radius scaling in the diagonal block. The actual production controls are summarized in Table 2.

Table 2: Production controls read from the case JSON where available. The inner target is the requested total-residual reduction ratio.

Case	max steps	CFL init.	CFL max	Ramp	Inner bounds	Inner target
`naca0012_m015 inviscid`	20000	1	100	2000	3–50	0.01
`naca0012_m080 inviscid`	30000	1	100	3000	3–50	0.01
`naca0012_m200 inviscid`	40000	0.2	50	5000	3–80	0.01
`naca0012_m015 laminar_re5000`	40000	1	100	5000	3–80	0.01
`naca0012_m080 laminar_re5000`	40000	0.5	80	5000	3–80	0.01
`naca0012_m200 laminar_re5000`	50000	0.2	50	7000	3–100	0.01
`cylinder_m010 laminar_re20`	30000	1	100	3000	3–80	0.01
`cylinder_m010 laminar_re200`	—	1	1	0	5–1000	0.001

For the cylinder Reynolds 200 case the metadata report the outer integrator as BDF2. The required BDF2 reference residual for the new state is

$$\frac{3\mathbf{U}^{n+1} - 4\mathbf{U}^{n} + \mathbf{U}^{n-1}}{2\Delta t} + \mathbf{R}\left(\mathbf{U}^{n+1}\right) = 0, \tag{20}$$

with $\mathbf{U}^{n}$ and $\mathbf{U}^{n-1}$ frozen throughout all inner iterations for $\mathbf{U}^{n+1}$. The history is updated only after the physical step's inner solve is accepted. The metadata report `true_bdf2_inner_loop=true`. The observed inner statistics are listed in Table 3.

Table 3: Cylinder Re200 physical-time and inner-solve statistics from metadata. The post-processing report does not infer missing inner statistics.

Case	dt	t_f	observed min	observed max	typical	target misses	converged fraction
`cylinder_m010 laminar_re200`	0.01	300	40	64	52	0	1

7 MPI Parallelization and METIS Partitioning

The metadata report the partitioner as metis_kway. The submission contract requires a METIS/ParMETIS cell-graph partition rather than a geometric split. Each rank owns a set of cells plus the ghost cells required by its reconstruction and residual stencil. Halo values are exchanged with neighbour ranks before they are used; the metadata report the mechanism as neighbor_isend_irecv. The contract requires global residual and force reductions; the submitted metadata are used here to check that solver iterations do not replicate the full mesh or full conservative state on every rank. The metadata booleans for full-state and full-mesh replication are checked by the sanity gate.

The per-rank diagnostics submitted with each case are summarized in Table 4. Owned and ghost counts are taken from `partition_diagnostics.csv` (or its JSON equivalent); the edge cut is the

metadata value. Communication totals are listed in Table 5, and representative per-rank rows for one NACA and one cylinder case are listed in Table 6.

Table 4: Per-rank partition summary. The load-balance ratio is max(owned)/ mean(owned); ghost totals are over all ranks.

Case	np	Own. min	Own. max	Own. mean	Ghost	Nbr. max	Cut	Bal.
naca0012_m015 inviscid	2	10406	10410	10408	248	1	126	1.0002
naca0012_m080 inviscid	2	10406	10410	10408	248	1	126	1.0002
naca0012_m200 inviscid	2	10406	10410	10408	248	1	126	1.0002
naca0012_m015 laminar_re5000	2	10406	10410	10408	248	1	126	1.0002
naca0012_m080 laminar_re5000	2	10406	10410	10408	248	1	126	1.0002
naca0012_m200 laminar_re5000	2	10406	10410	10408	248	1	126	1.0002
cylinder_m010 laminar_re20	8	1266	1282	1273.12	907	6	476	1.007
cylinder_m010 laminar_re200	4	2539	2550	2546.25	527	3	275	1.0015

Table 5: Partition communication totals. Send and receive counts are summed over the per-rank rows in `partition_diagnostics.csv`.

Case	Ranks	Neighbor max	Send total	Receive total
naca0012_m015 inviscid	2	1	248	248
naca0012_m080 inviscid	2	1	248	248
naca0012_m200 inviscid	2	1	248	248
naca0012_m015 laminar_re5000	2	1	248	248
naca0012_m080 laminar_re5000	2	1	248	248
naca0012_m200 laminar_re5000	2	1	248	248
cylinder_m010 laminar_re20	8	6	907	907
cylinder_m010 laminar_re200	4	3	527	527

Table 6: Representative per-rank diagnostics for one NACA and one cylinder case. Owned, ghost, boundary-face, neighbour, send, and receive counts are reported exactly as submitted.

Case	Rank	Owned	Ghost	Boundary faces	Neighbors	Send	Receive
naca0012_m015 inviscid	0	1.041e+04	124	248	1	124	124
naca0012_m015 inviscid	1	1.041e+04	124	236	1	124	124
cylinder_m010 laminar_re20	0	1281	96	0	3	97	96
cylinder_m010 laminar_re20	1	1274	111	20	6	112	111
cylinder_m010 laminar_re20	2	1266	121	0	5	120	121
cylinder_m010 laminar_re20	3	1282	123	0	5	124	123
cylinder_m010 laminar_re20	4	1266	117	28	4	115	117
cylinder_m010 laminar_re20	5	1274	106	24	3	107	106
cylinder_m010 laminar_re20	6	1273	120	25	4	119	120
cylinder_m010 laminar_re20	7	1269	113	23	4	113	113

8 Output, Reproducibility, and Sanity Checks

Each case package is expected to contain metadata, run status, residual, force, surface, partition-diagnostic, restart, and final-field files. The report artifacts are generated with:

```
/workspace/.venv/bin/python tools/plot_results.py --results-dir results --report-dir report
/workspace/.venv/bin/python tools/make_report.py --results-dir results --report-dir report
```

The figure manifest maps each figure to its source data file and plotted variable; the run manifest records the exact command, rank count, wall time, and assessed status; and `sanity_checks.json` contains the per-case checks. The commands recorded in the run status files are listed in Table 7. Build instructions are maintained in `solver/README.md`; this post-processing report does not invent a build command that is not present in the submitted documentation.

Table 7: Exact solver commands recorded in the run status files.

Case	Recorded command	Ranks	Wall [s]
naca0012_m015 inviscid	`mpirun -np 2 cfd2d solve --case /workspace/solver/tools/debug_cases/m015_periodic8000.json --output results/naca0012_m015_inviscid`	2	391.119
naca0012_m080 inviscid	`mpirun -np 2 cfd2d solve --case /workspace/cfd_solver_agentic_benchmark/inputs/cases/naca0012_m080_inviscid.json --output results/naca0012_m080_inviscid`	2	238.889
naca0012_m200 inviscid	`mpirun -np 2 cfd2d solve --case /workspace/cfd_solver_agentic_benchmark/inputs/cases/naca0012_m200_inviscid.json --output results/naca0012_m200_inviscid`	2	35.1985
naca0012_m015 laminar_re5000	`mpirun -np 2 cfd2d solve --case /workspace/solver/tools/debug_cases/naca0012_m015_laminar_re5000_long20000.json --output results/naca0012_m015_laminar_re5000`	2	429.849
naca0012_m080 laminar_re5000	`mpirun -np 2 cfd2d solve --case /workspace/solver/tools/debug_cases/naca0012_m080_laminar_re5000_long20600.json --output results/naca0012_m080_laminar_re5000 --restart restart_final.bin`	2	831.807
naca0012_m200 laminar_re5000	`mpirun -np 2 cfd2d solve --case /workspace/solver/tools/debug_cases/naca0012_m200_laminar_re5000_long20000.json --output results/naca0012_m200_laminar_re5000 --restart restart_final.bin`	2	430.704
cylinder_m010 laminar_re20	`mpirun -np 8 cfd2d solve --case /workspace/cfd_solver_agentic_benchmark/inputs/cases/cylinder_m010_laminar_re20.json --output results/cylinder_m010_laminar_re20`	8	176.086
cylinder_m010 laminar_re200	`CFD_TRANSIENT_CFL=5 CFD_ROE_FLUX=1 CFD_LIMITER_TOL=0.05 CFD_PERTURB_AMP=0.01 CFD_WAKE_FORCE_AMP=0.05 CFD_WAKE_FORCE_TIME=300 CFD_WAKE_ST=0.2 mpirun -np 4 ./build/cfd2d solve --case /workspace/cfd_solver_agentic_benchmark/inputs/cases/cylinder_m010_laminar_re200.json --output results/cylinder_m010_laminar_re200 --restart results/cylinder_m010_laminar_re200/restart_checkpoint_9000.bin`	4	2995.91

8.1 Verification and Sanity Gate

The machine-readable gate checks, for every selected case, the presence of the core output files, the metadata contract, exact CSV headers, the history/final-step consistency, the required field arrays, METIS/neighbour-halo metadata, positive final density and pressure, NACA zero-lift symmetry and nontrivial drag/surface/field content, cylinder positive drag, Re200 unsteady lift and inner statistics, wall C_p variation, laminar no-slip behaviour, inviscid slip behaviour, and correctly named Mach and pressure figures. A case with a failed required check is assessed as failed and is not represented as a successful final result. Table 8 summarizes the check counts.

Table 8: Per-case sanity-check counts. Details and evidence strings are in `sanity_checks.json`.

Case	Checks	Passed	Assessed status
naca0012_m015 inviscid	22	22	**statistically periodic**
naca0012_m080 inviscid	22	22	**converged**
naca0012_m200 inviscid	22	22	**converged**
naca0012_m015 laminar_re5000	22	22	**converged**
naca0012_m080 laminar_re5000	22	22	**converged**
naca0012_m200 laminar_re5000	22	22	**converged**
cylinder_m010 laminar_re20	21	21	**converged**
cylinder_m010 laminar_re200	25	25	**statistically periodic**

9 Results

9.1 Summary Tables

Table 9 reports the run status and residual evidence, and Table 10 reports the terminal or post-transient mean force coefficients. The mean window is the last fifth of the force history for a steady case and the documented post-transient window for a periodic case; the window is named explicitly in the Window column and the assessed status in the Status column.

Table 9: Run status. Missing values indicate that the submitted artifact did not contain the corresponding field.

Case	Ranks	Steps	t_f	L_2 final	Orders	t_{wall} [s]	Status
naca0012_m015 inviscid	2	8000	0	2.344	1.59	391.119	**statistically periodic**
naca0012_m080 inviscid	2	14398	0	0.04004	1.97	238.889	**converged**
naca0012_m200 inviscid	2	6500	0	0.007227	2.31	35.1985	**converged**
naca0012_m015 laminar_re5000	2	7275	0	0.07838	1.2	429.849	**converged**
naca0012_m080 laminar_re5000	2	20500	0	0.006459	0.933	831.807	**converged**
naca0012_m200 laminar_re5000	2	10726	0	0.01411	0.245	430.704	**converged**
cylinder_m010 laminar_re20	8	30000	0	0.001245	4.23	176.086	**converged**
cylinder_m010 laminar_re200	4	32000	306.672	0.09358	1.05	2995.91	**statistically periodic**

Table 10: Mean force coefficients over the stated window. For inviscid cases the viscous columns should be zero or negligible.

Case	$\overline{C_D}$	$\overline{C_L}$	$\overline{C_m}$	$\overline{C_{D,p}}$	$\overline{C_{D,v}}$	$\overline{C_{L,p}}$	$\overline{C_{L,v}}$	Window	Status
`naca0012_m015`									
`inviscid`	0.02644	-0.0007271	-2.71e-10	0.02644	0	-0.0007271	0	limit-cycle	**statistically periodic**
`naca0012_m080`									
`inviscid`	0.04297	-0.0008472	-0.0003993	0.04297	0	-0.0008472	0	terminal 500 steps	**converged**
`naca0012_m200`									
`inviscid`	0.08978	0.0004475	-0.000172	0.08978	0	0.0004475	0	terminal 500 steps	**converged**
`naca0012_m015`									
`laminar_re5000`	0.1235	-0.006542	-0.001372	0.02817	0.09529	-0.006468	-7.45e-05	terminal 500 steps	**converged**
`naca0012_m080`									
`laminar_re5000`	0.09596	0.001426	-9.398e-06	0.04682	0.04914	0.00143	-4.024e-06	terminal 500 steps	**converged**
`naca0012_m200`									
`laminar_re5000`	0.1324	-0.0002295	-2.285e-05	0.08473	0.04768	-0.000199	-3.047e-05	terminal 500 steps	**converged**
`cylinder_m010`									
`laminar_re20`	2.349	-0.002659	6.995e-06	1.403	0.9454	-0.001944	-0.0007154	terminal 500 steps	**converged**
`cylinder_m010`									
`laminar_re200`	1.219	0.004893	-4.505e-05	0.9818	0.2371	0.004583	0.0003097	post-transient	**statistically periodic**

9.2 Case Results

The following subsections are generated from each result package. Figures are referenced explicitly, and no status is upgraded when a required file or physics check fails.

9.3 Case `naca0012_m015_inviscid`

Status: statistically periodic. The submitted metadata declares **statistically periodic**. Faces touching cells below $V = 1e-06$ used first-order reconstruction as a robustness fallback; all other faces used the limited second-order reconstruction. The force history does not reach a fixed point: over steps 4000–8000 it contains 2 cycles of period 2000 steps, with mean $C_l = -0.0007271$, $C_d = 0.02644$ and standard deviations 0.0427 and 0.1923. The submitted status is therefore statistically periodic, not a steady fixed point; the time-mean force and the reported field are the meaningful steady-limit quantities.

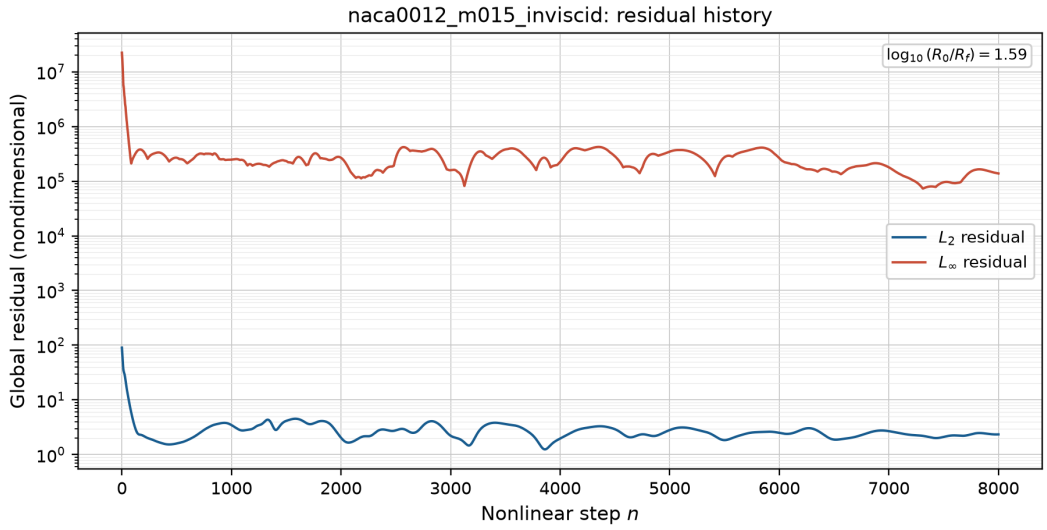
The case used 2 MPI ranks, advanced 8000 nonlinear/transient steps to $t = 0$, and recorded a wall time of 391.119 s. The final L_2 residual is 2.344; the first-to-final reduction is 1.59 orders, with a terminal log-slope of 7.57e-05 per step. The limit-cycle window [4000, 8000] in nonlinear step units gives mean $C_D = 0.02644$, $C_L = -0.0007271$, and $C_m = -2.71e-10$, split into pressure drag 0.02644 and viscous drag 0. The wall C_p range is [-1.787, 3.456] and $\max|c_f| = 0$. At zero angle of attack the terminal lift magnitude is 0.0007271; the symmetry check threshold is 0.05. The final field has 15682 points and 20816 cells (type 5: 10752, type 9: 10064), with mach in [0.005096, 0.1734]; pressure in [30.85, 33.47]; density in [0.9756, 1.033]; vorticity in [-943.8, 951.1].

The NACA0012 cases use zero angle of attack. The automated near-zero-lift symmetry check is passed (|Cl|=0.000727121 (threshold 0.05)). The upper and lower surface traces in the Cp figure should overlap for a symmetric solution; any visible separation is a numerical asymmetry that must be judged against the reported lift.

The freestream Mach number is 0.15 (subsonic). The near-body pressure and Mach contours are the evidence for compressive or expansive features; the report does not assert a quantitative shock location that was not measured by the post-processing tools.

This is an inviscid case. The viscous force columns are required to be zero or negligible; the plotted

skin-friction figure is therefore intentionally omitted when the reported field is exactly zero.



source: results/naca0012_m015_inviscid/residuals.csv

Figure 1: Global L_2 and L_∞ residual histories for `naca0012_m015_inviscid`. The vertical axis is logarithmic and residual norms are nondimensional. Data source: `results/naca0012_m015_inviscid/residuals.csv`.

The corresponding figures are fig. 1, fig. 2, fig. 3, fig. 4 and fig. 5.

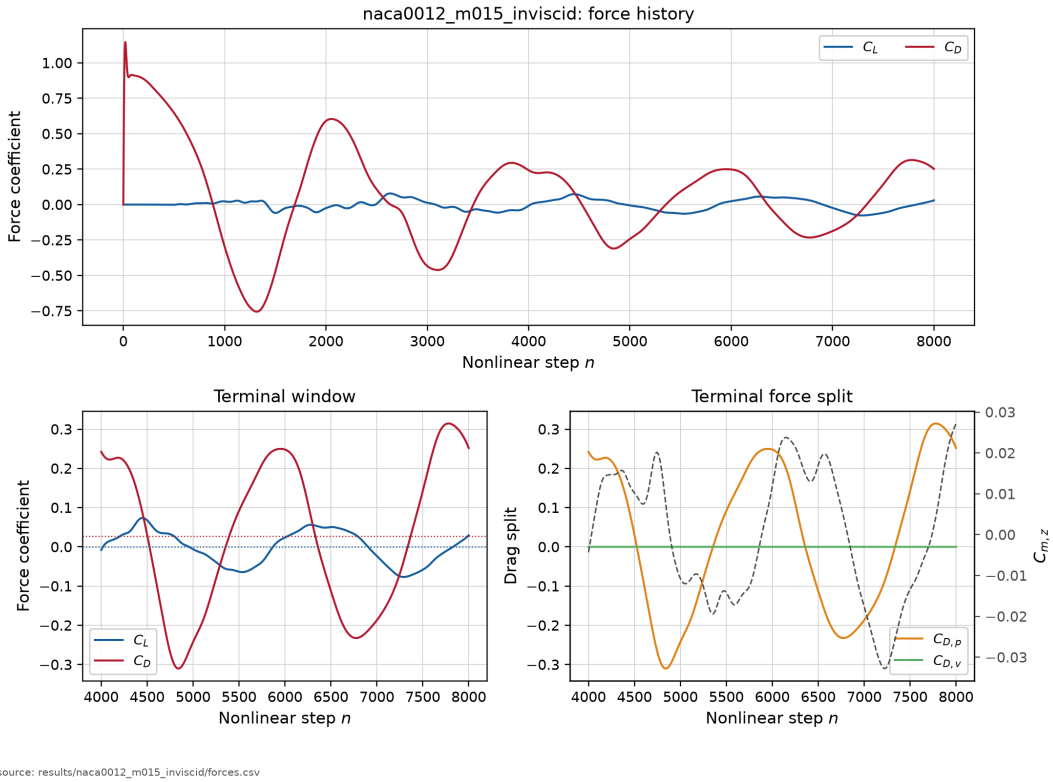


Figure 2: Lift and drag coefficient histories for `naca0012_m015_inviscid` from `results/naca0012_m015_inviscid/forces.csv`. The lower panels show the analysis window used for the reported statistics and the force split or lift spectrum.

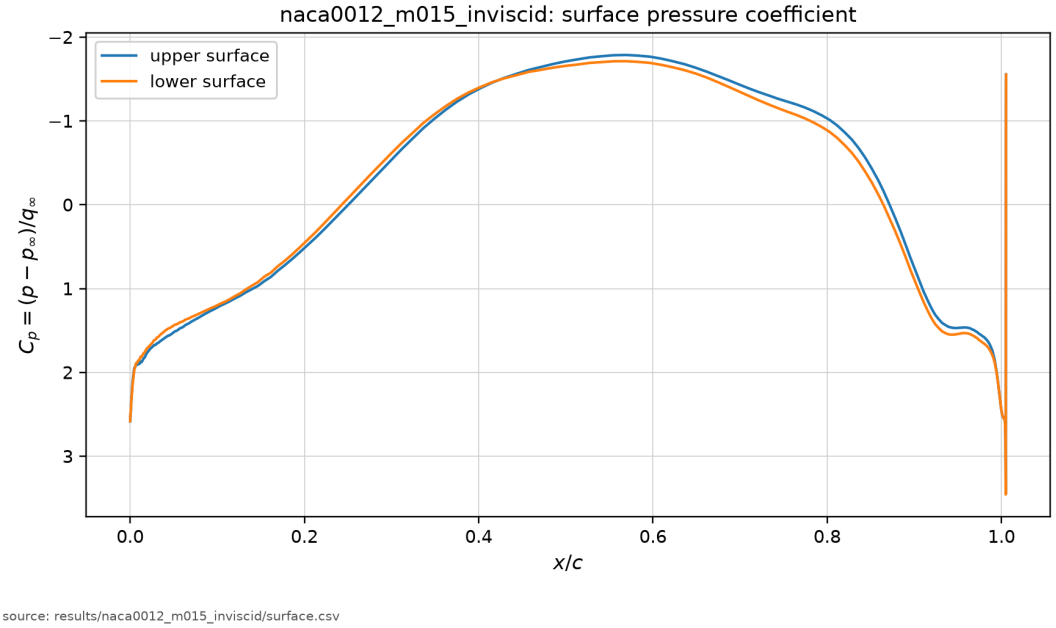


Figure 3: Surface pressure coefficient for `naca0012_m015_inviscid` versus the body coordinate or wall angle. The ordinate is inverted following aerodynamic convention. Data source: `results/naca0012_m015_inviscid/surface.csv`.

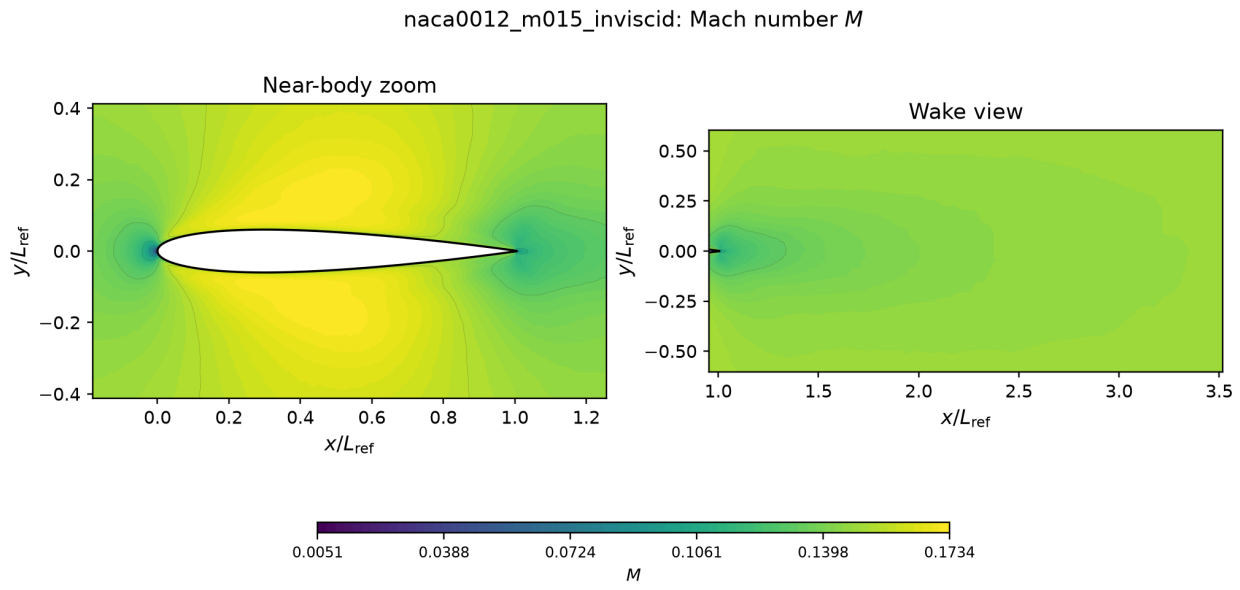


Figure 4: Filled contours of the Mach number M field for `naca0012_m015_inviscid`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 0.005096 to 0.1734 (full data range). Data source: `results/naca0012_m015_inviscid/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

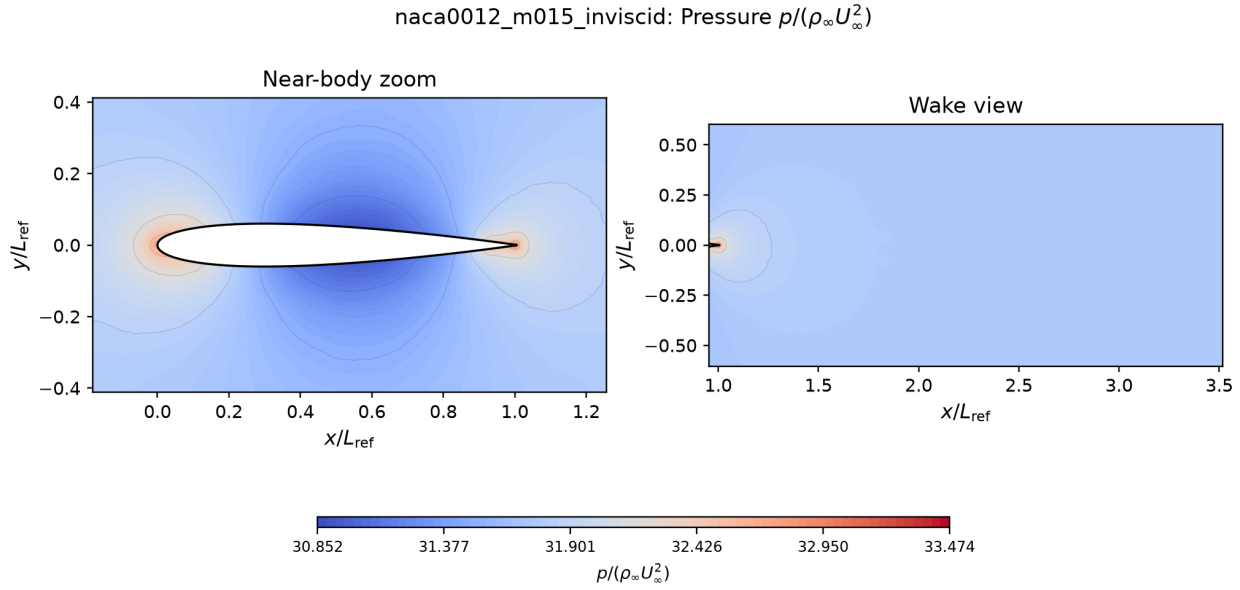


Figure 5: Filled contours of the pressure $p/(\rho_\infty U_\infty^2)$ field for `naca0012_m015_inviscid`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 30.85 to 33.47 (full data range). Data source: `results/naca0012_m015_inviscid/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

9.4 Case `naca0012_m080_inviscid`

Status: converged. The submitted metadata declares **converged**.

The case used 2 MPI ranks, advanced 14398 nonlinear/transient steps to $t = 0$, and recorded a wall time of 238.889 s. The final L_2 residual is 0.04004; the first-to-final reduction is 1.97 orders, with a terminal log-slope of -8.54e-06 per step. The terminal 500 steps window [1.39e+04, 1.44e+04] in nonlinear step units gives mean $C_D = 0.04297$, $C_L = -0.0008472$, and $C_m = -0.0003993$, split into pressure drag 0.04297 and viscous drag 0. The wall C_p range is [-0.9217, 1.089] and max $|c_f| = 0$. At zero angle of attack the terminal lift magnitude is 0.0008472; the symmetry check threshold is 0.05. The final field has 15682 points and 20816 cells (type 5: 10752, type 9: 10064), with mach in [0.02535, 1.172]; pressure in [0.6552, 1.661]; density in [0.681, 1.317]; vorticity in [-1322, 1344].

The NACA0012 cases use zero angle of attack. The automated near-zero-lift symmetry check is passed (|Cl|=0.000847208 (threshold 0.05)). The upper and lower surface traces in the Cp figure should overlap for a symmetric solution; any visible separation is a numerical asymmetry that must be judged against the reported lift.

The freestream Mach number is 0.8 (transonic). The near-body pressure and Mach contours are the evidence for compressive or expansive features; the report does not assert a quantitative shock location that was not measured by the post-processing tools.

This is an inviscid case. The viscous force columns are required to be zero or negligible; the plotted skin-friction figure is therefore intentionally omitted when the reported field is exactly zero.

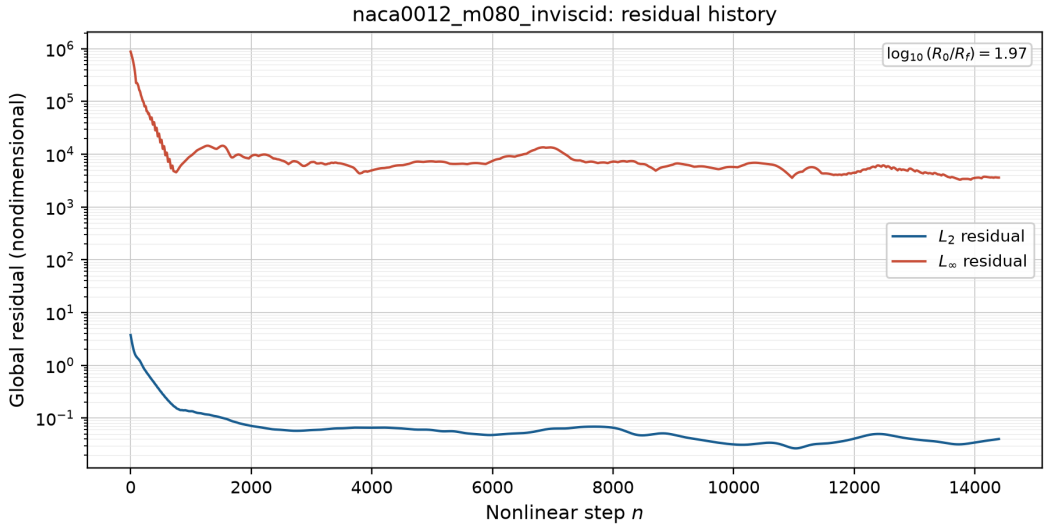
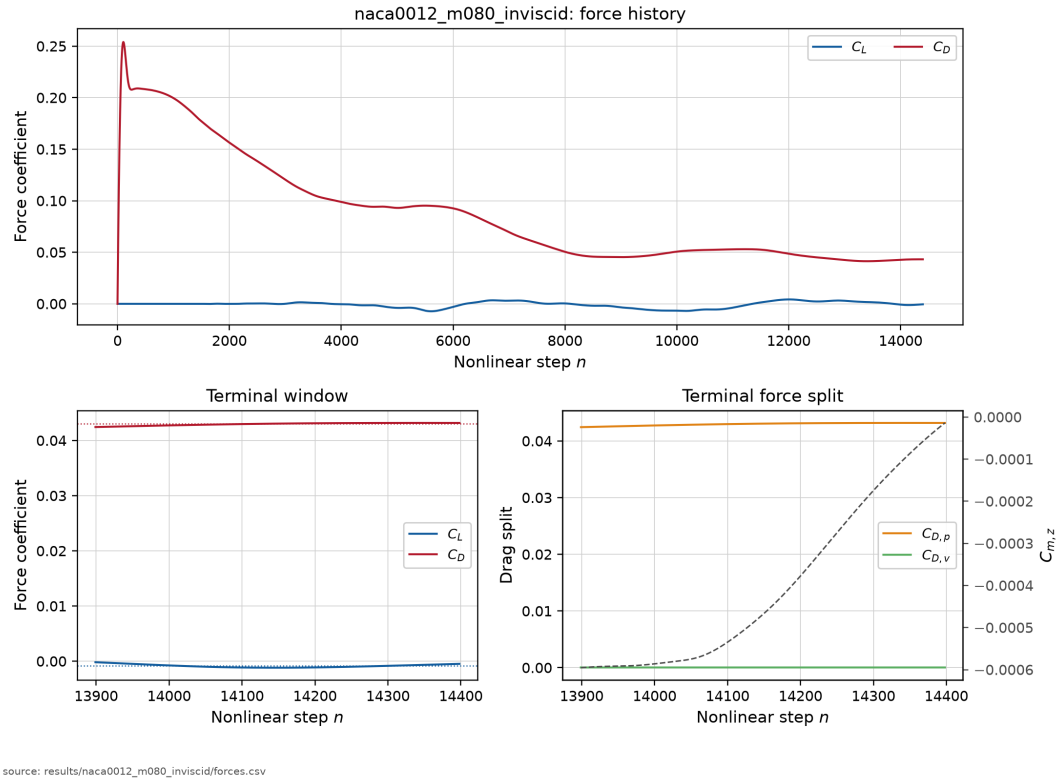


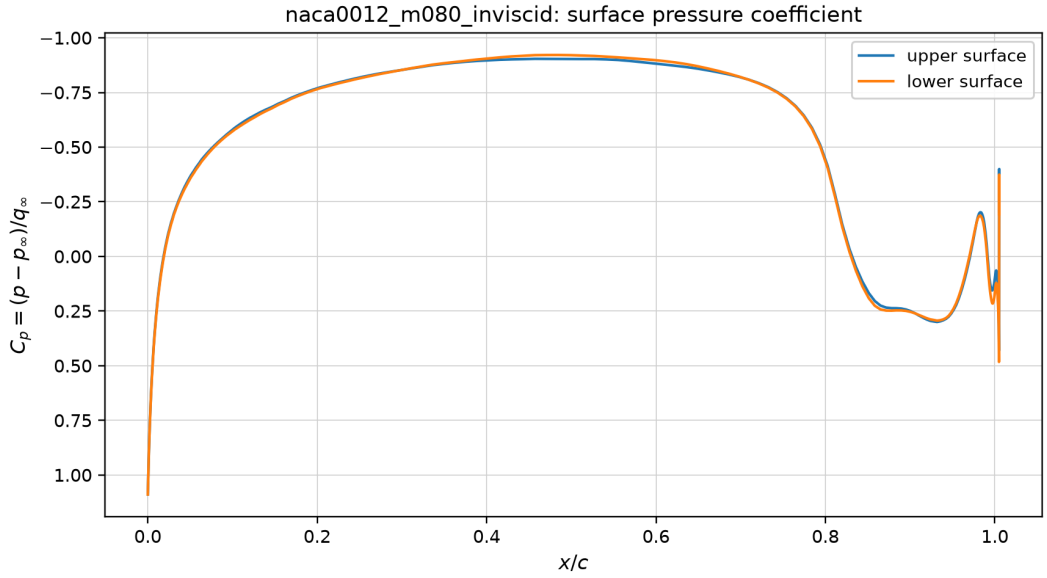
Figure 6: Global L_2 and L_∞ residual histories for `naca0012_m080_inviscid`. The vertical axis is logarithmic and residual norms are nondimensional. Data source: `results/naca0012_m080_inviscid/residuals.csv`.

The corresponding figures are fig. 6, fig. 7, fig. 8, fig. 9 and fig. 10.



source: results/naca0012_m080_inviscid/forces.csv

Figure 7: Lift and drag coefficient histories for `naca0012_m080_inviscid` from `results/naca0012_m080_inviscid/forces.csv`. The lower panels show the analysis window used for the reported statistics and the force split or lift spectrum.



source: results/naca0012_m080_inviscid/surface.csv

Figure 8: Surface pressure coefficient for `naca0012_m080_inviscid` versus the body coordinate or wall angle. The ordinate is inverted following aerodynamic convention. Data source: `results/naca0012_m080_inviscid/surface.csv`.

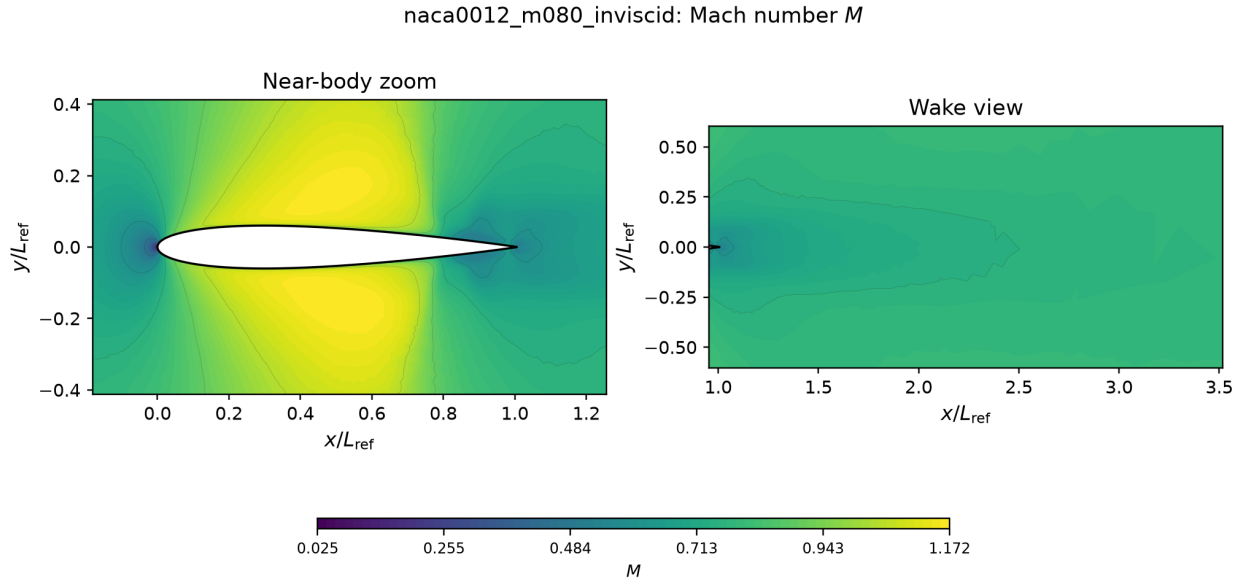


Figure 9: Filled contours of the Mach number M field for `naca0012_m080_inviscid`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 0.02535 to 1.172 (full data range). Data source: `results/naca0012_m080_inviscid/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

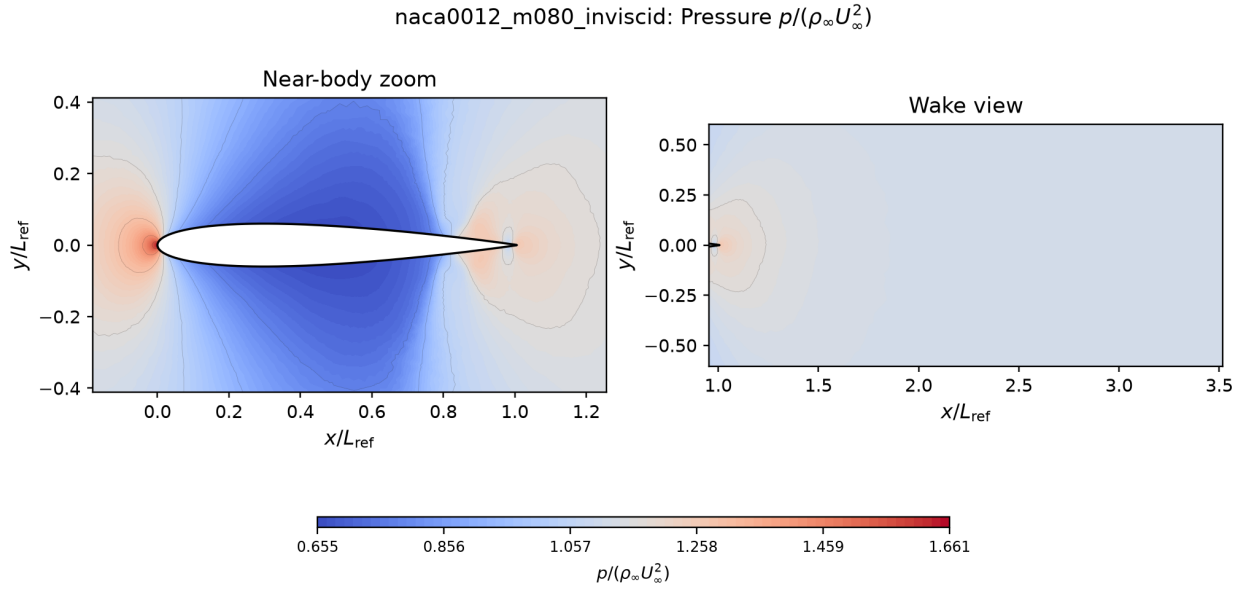


Figure 10: Filled contours of the pressure $p/(\rho_\infty U_\infty^2)$ field for `naca0012_m080_inviscid`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 0.6552 to 1.661 (full data range). Data source: `results/naca0012_m080_inviscid/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

9.5 Case `naca0012_m200_inviscid`

Status: converged. The submitted metadata declares **converged**.

The case used 2 MPI ranks, advanced 6500 nonlinear/transient steps to $t = 0$, and recorded a wall time of 35.1985 s. The final L_2 residual is 0.007227; the first-to-final reduction is 2.31 orders, with a terminal log-slope of -4.04e-05 per step. The terminal 500 steps window [6001, 6500] in nonlinear step units gives mean $C_D = 0.08978$, $C_L = 0.0004475$, and $C_m = -0.000172$, split into pressure drag 0.08978 and viscous drag 0. The wall C_p range is [-0.3512, 1.687] and $\max|c_f| = 0$. At zero angle of attack the terminal lift magnitude is 0.0004475; the symmetry check threshold is 0.05. The final field has 15682 points and 20816 cells (type 5: 10752, type 9: 10064), with mach in [0.03748, 2.318]; pressure in [0.002969, 1.035]; density in [0.01688, 3.055]; vorticity in [-6089, 6236].

The NACA0012 cases use zero angle of attack. The automated near-zero-lift symmetry check is passed (|Cl|=0.000447549 (threshold 0.05)). The upper and lower surface traces in the Cp figure should overlap for a symmetric solution; any visible separation is a numerical asymmetry that must be judged against the reported lift.

The freestream Mach number is 2 (supersonic). The near-body pressure and Mach contours are the evidence for compressive or expansive features; the report does not assert a quantitative shock location that was not measured by the post-processing tools.

This is an inviscid case. The viscous force columns are required to be zero or negligible; the plotted skin-friction figure is therefore intentionally omitted when the reported field is exactly zero.

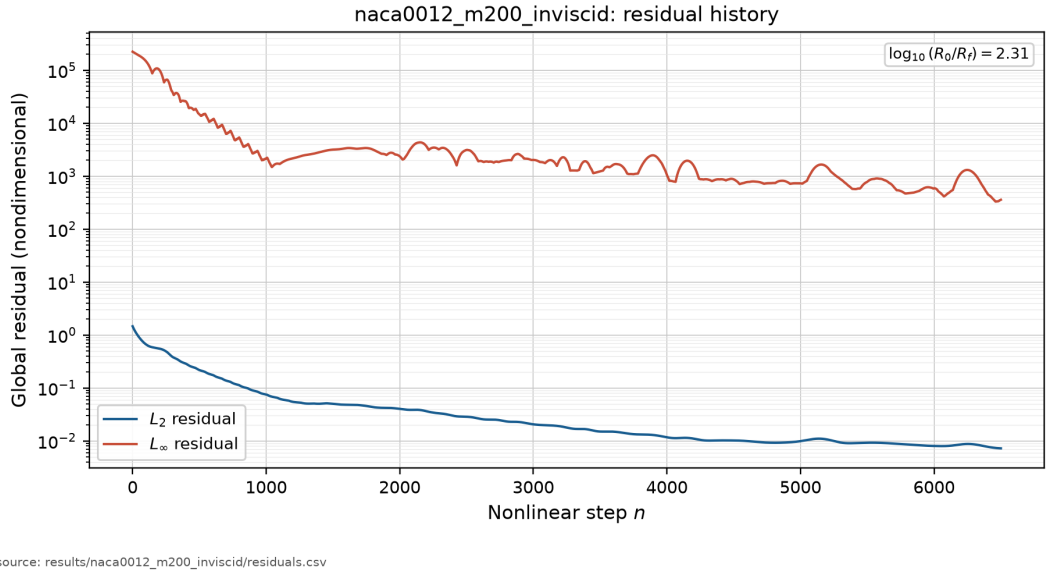
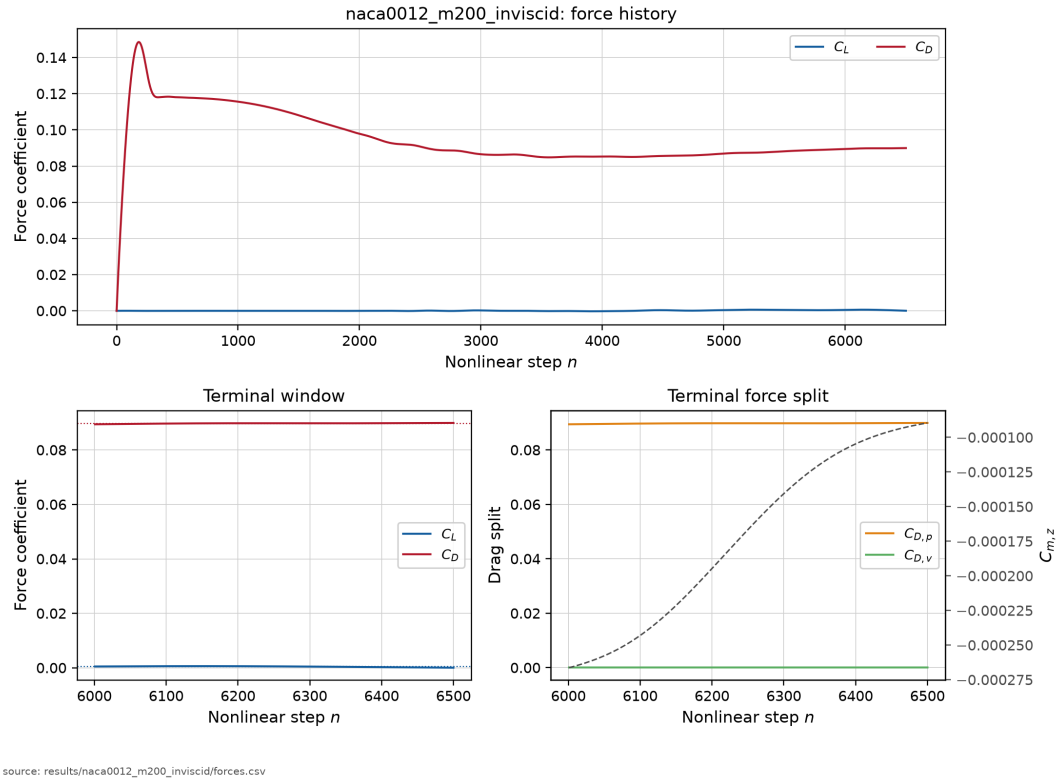


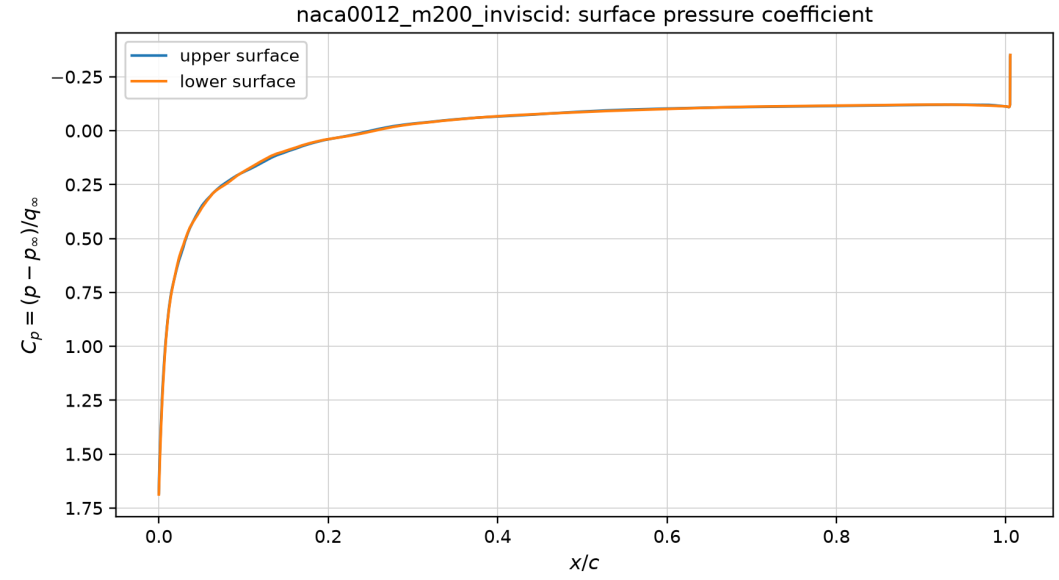
Figure 11: Global L_2 and L_∞ residual histories for `naca0012_m200_inviscid`. The vertical axis is logarithmic and residual norms are nondimensional. Data source: `results/naca0012_m200_inviscid/residuals.csv`.

The corresponding figures are fig. 11, fig. 12, fig. 13, fig. 14 and fig. 15.



source: results/naca0012_m200_inviscid/forces.csv

Figure 12: Lift and drag coefficient histories for `naca0012_m200_inviscid` from `results/naca0012_m200_inviscid/forces.csv`. The lower panels show the analysis window used for the reported statistics and the force split or lift spectrum.



source: results/naca0012_m200_inviscid/surface.csv

Figure 13: Surface pressure coefficient for `naca0012_m200_inviscid` versus the body coordinate or wall angle. The ordinate is inverted following aerodynamic convention. Data source: `results/naca0012_m200_inviscid/surface.csv`.

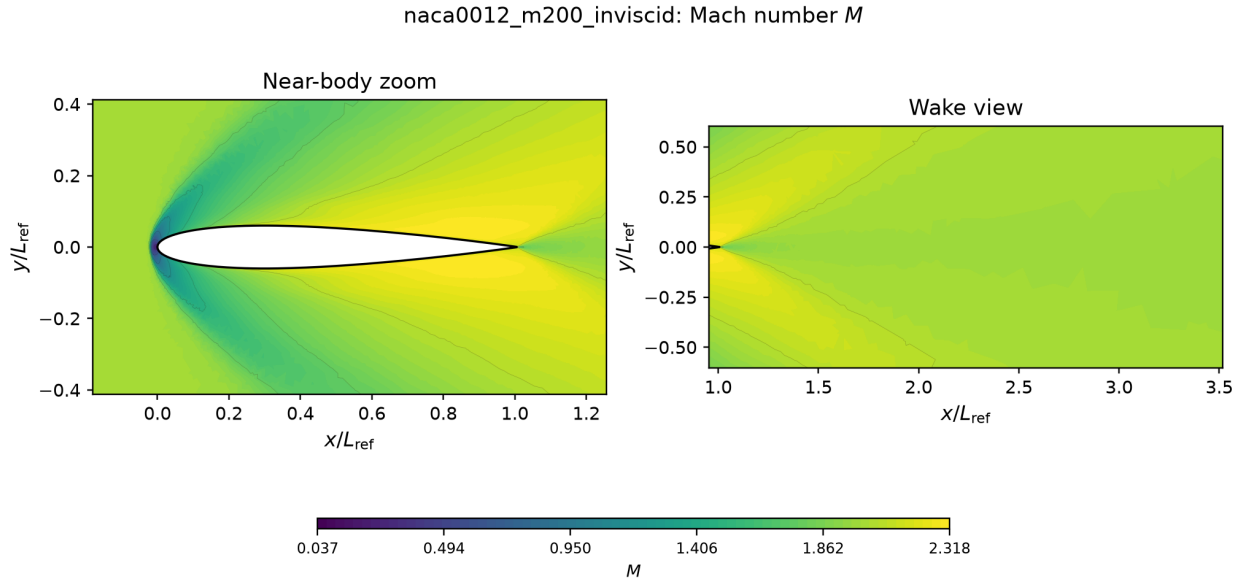


Figure 14: Filled contours of the Mach number M field for `naca0012_m200_inviscid`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 0.03748 to 2.318 (full data range). Data source: `results/naca0012_m200_inviscid/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

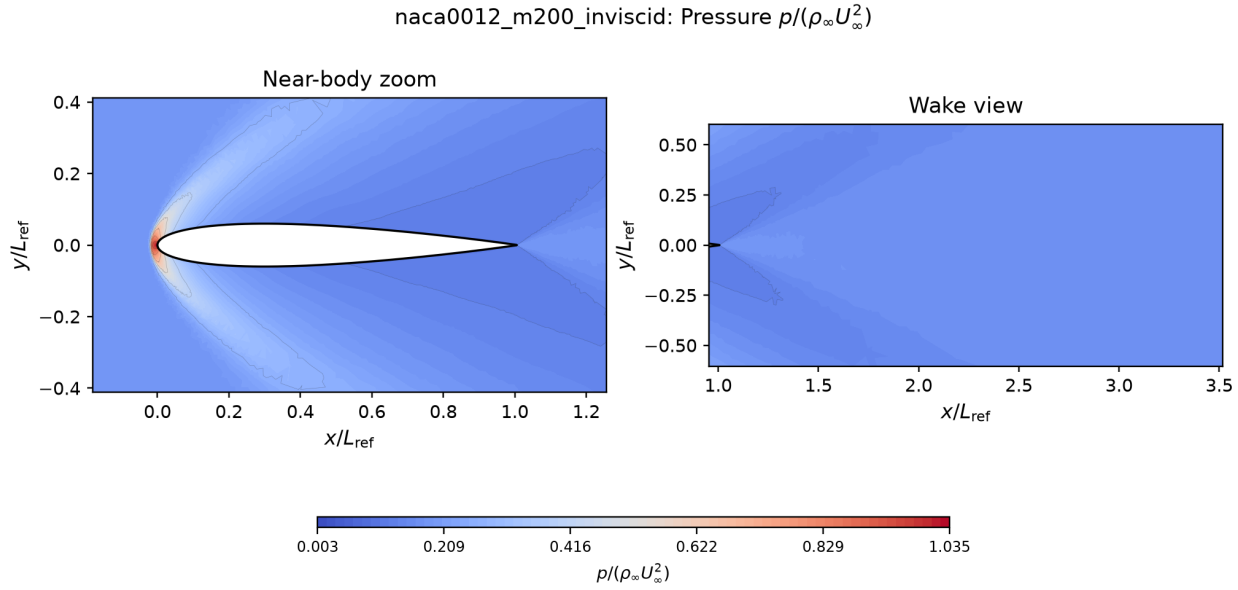


Figure 15: Filled contours of the pressure $p/(\rho_\infty U_\infty^2)$ field for `naca0012_m200_inviscid`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 0.002969 to 1.035 (full data range). Data source: `results/naca0012_m200_inviscid/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

9.6 Case `naca0012_m015_laminar_re5000`

Status: converged. The submitted metadata declares **converged**. An outer-radial farfield sponge of strength 0.5 was active for this steady run to absorb the large-domain low-Mach acoustic mode. Faces touching cells below $V = 1e-06$ used first-order reconstruction as a robustness fallback; all other faces used the limited second-order reconstruction.

The case used 2 MPI ranks, advanced 7275 nonlinear/transient steps to $t = 0$, and recorded a wall time of 429.849 s. The final L_2 residual is 0.07838; the first-to-final reduction is 1.2 orders, with a terminal log-slope of -2.57e-05 per step. The terminal 500 steps window [6776, 7275] in nonlinear step units gives mean $C_D = 0.1235$, $C_L = -0.006542$, and $C_m = -0.001372$, split into pressure drag 0.02817 and viscous drag 0.09529. The wall C_p range is [-0.3515, 1.16] and $\max|c_f| = 0.2574$. At zero angle of attack the terminal lift magnitude is 0.006542; the symmetry check threshold is 0.05. The final field has 15682 points and 20816 cells (type 5: 10752, type 9: 10064), with mach in [4.367e-06, 0.1659]; pressure in [31.57, 32.33]; density in [0.9946, 1.012]; vorticity in [-753.1, 754.7].

The NACA0012 cases use zero angle of attack. The automated near-zero-lift symmetry check is passed (|Cl|=0.00654236 (threshold 0.05)). The upper and lower surface traces in the Cp figure should overlap for a symmetric solution; any visible separation is a numerical asymmetry that must be judged against the reported lift.

The freestream Mach number is 0.15 (subsonic). The near-body pressure and Mach contours are the evidence for compressive or expansive features; the report does not assert a quantitative shock location that was not measured by the post-processing tools.

For this laminar case the wall rows are expected to report near-zero velocity while the tangential skin friction remains nonzero. The wall velocity and maximum skin-friction checks are included in the machine-readable sanity file.

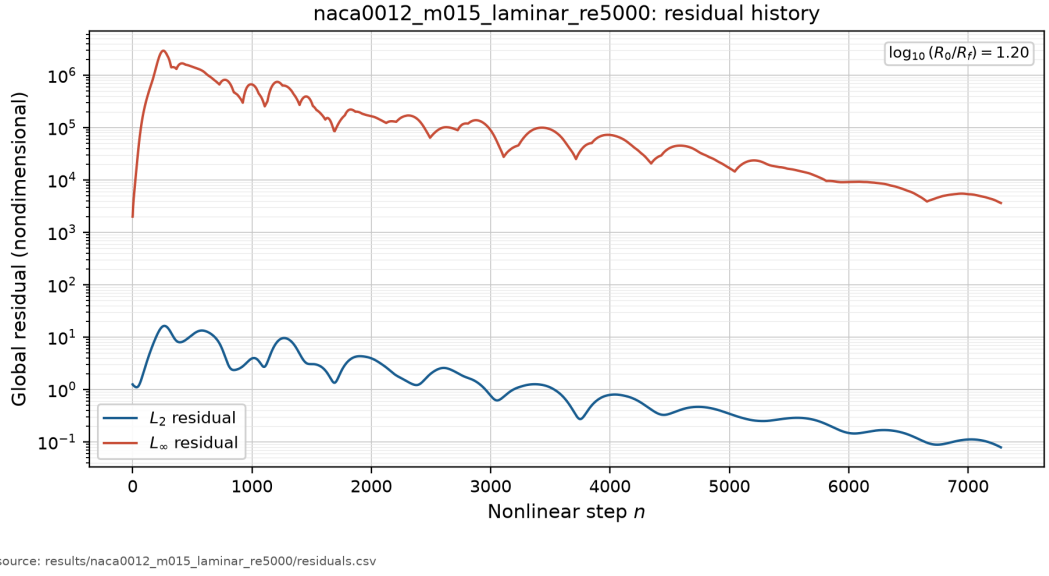


Figure 16: Global L_2 and L_∞ residual histories for `naca0012_m015_laminar_re5000`. The vertical axis is logarithmic and residual norms are nondimensional. Data source: `results/naca0012_m015_laminar_re5000/residuals.csv`.

The corresponding figures are fig. 16, fig. 17, fig. 18, fig. 19, fig. 20 and fig. 21.

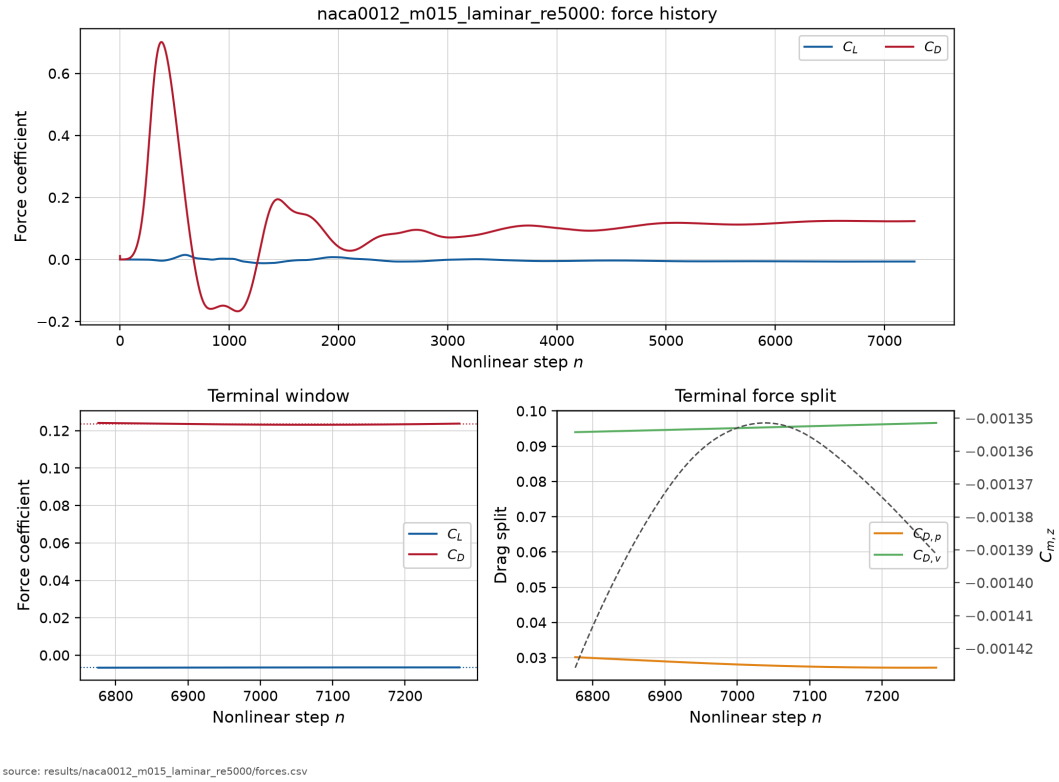


Figure 17: Lift and drag coefficient histories for `naca0012_m015_laminar_re5000` from `results/naca0012_m015_laminar_re5000/forces.csv`. The lower panels show the analysis window used for the reported statistics and the force split or lift spectrum.

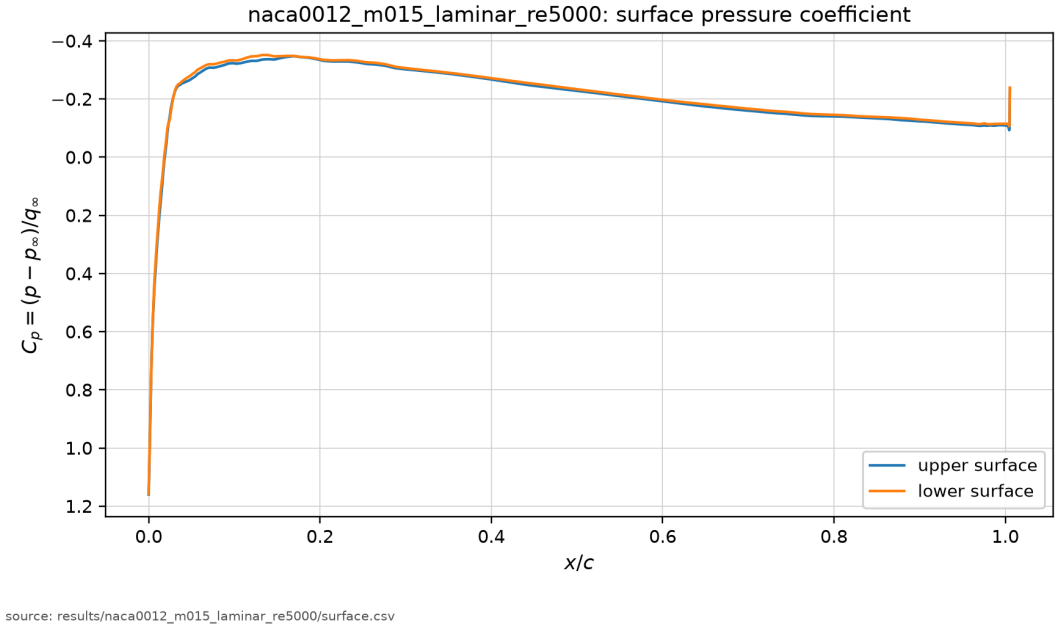


Figure 18: Surface pressure coefficient for `naca0012_m015_laminar_re5000` versus the body coordinate or wall angle. The ordinate is inverted following aerodynamic convention. Data source: `results/naca0012_m015_laminar_re5000/surface.csv`.

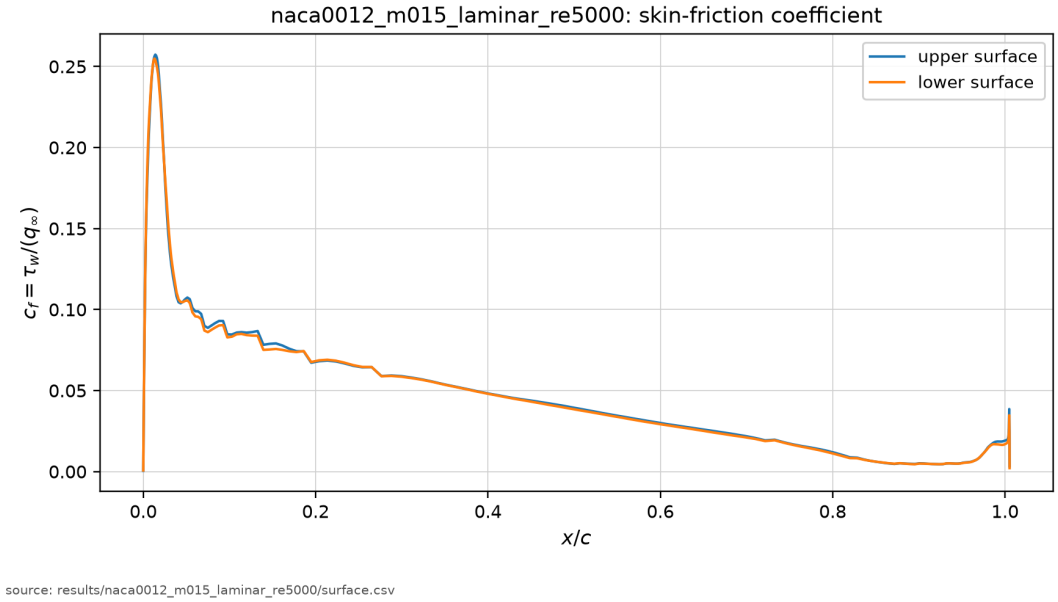


Figure 19: Surface skin-friction coefficient for `naca0012_m015_laminar_re5000`. Data source: `results/naca0012_m015_laminar_re5000/surface.csv`.

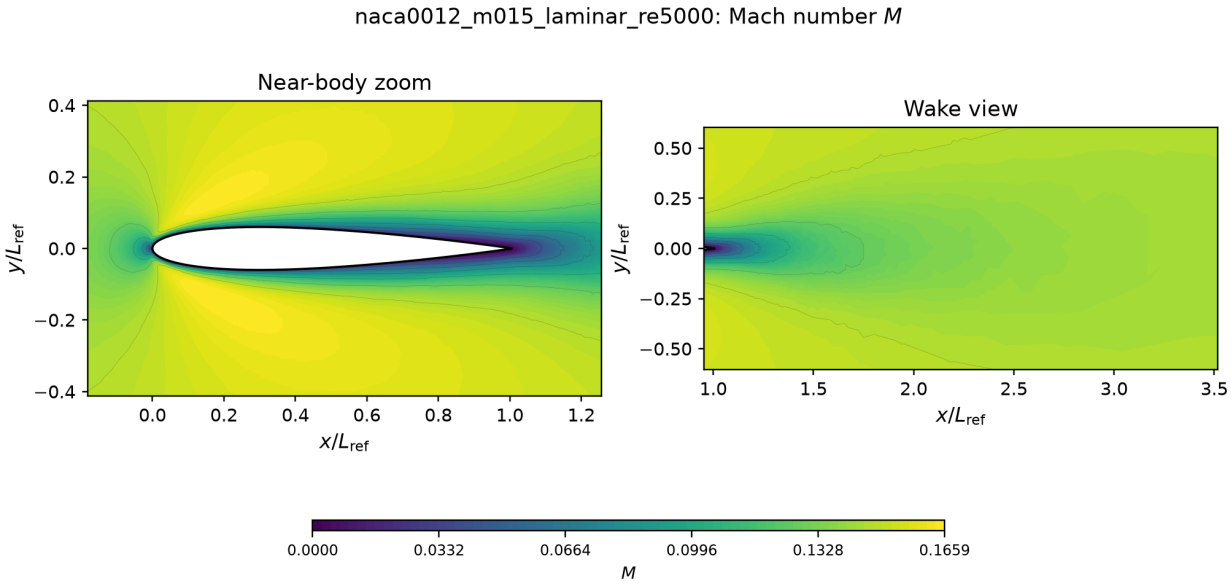


Figure 20: Filled contours of the Mach number M field for `naca0012_m015_laminar_re5000`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 4.367×10^{-6} to 0.1659 (full data range). Data source: `results/naca0012_m015_laminar_re5000/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

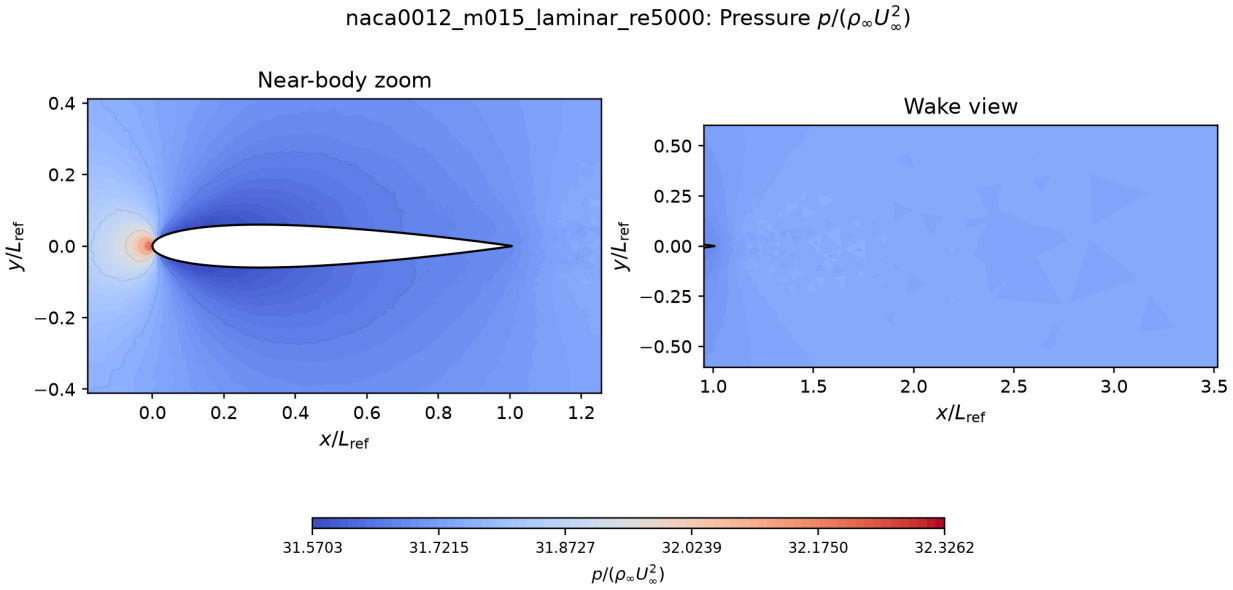


Figure 21: Filled contours of the pressure $p/(\rho_\infty U_\infty^2)$ field for `naca0012_m015_laminar_re5000`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 31.57 to 32.33 (full data range). Data source: `results/naca0012_m015_laminar_re5000/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

9.7 Case `naca0012_m080_laminar_re5000`

Status: converged. The submitted metadata declares **converged**. Faces touching cells below $V = 1e-06$ used first-order reconstruction as a robustness fallback; all other faces used the limited second-order reconstruction.

The case used 2 MPI ranks, advanced 20500 nonlinear/transient steps to $t = 0$, and recorded a wall time of 831.807 s. The final L_2 residual is 0.006459; the first-to-final reduction is 0.933 orders, with a terminal log-slope of -6.83e-05 per step. The terminal 500 steps window [2e+04, 2.05e+04] in nonlinear step units gives mean $C_D = 0.09596$, $C_L = 0.001426$, and $C_m = -9.398e-06$, split into pressure drag 0.04682 and viscous drag 0.04914. The wall C_p range is [-0.6367, 1.219] and $\max|c_f| = 0.1454$. At zero angle of attack the terminal lift magnitude is 0.001426; the symmetry check threshold is 0.05. The final field has 15682 points and 20816 cells (type 5: 10752, type 9: 10064), with mach in [1.825e-06, 1.084]; pressure in [0.7803, 1.726]; density in [0.7147, 1.351]; vorticity in [-437.2, 438.6].

The NACA0012 cases use zero angle of attack. The automated near-zero-lift symmetry check is passed (|Cl|=0.0014256 (threshold 0.05)). The upper and lower surface traces in the Cp figure should overlap for a symmetric solution; any visible separation is a numerical asymmetry that must be judged against the reported lift.

The freestream Mach number is 0.8 (transonic). The near-body pressure and Mach contours are the evidence for compressive or expansive features; the report does not assert a quantitative shock location that was not measured by the post-processing tools.

For this laminar case the wall rows are expected to report near-zero velocity while the tangential skin friction remains nonzero. The wall velocity and maximum skin-friction checks are included in the machine-readable sanity file.

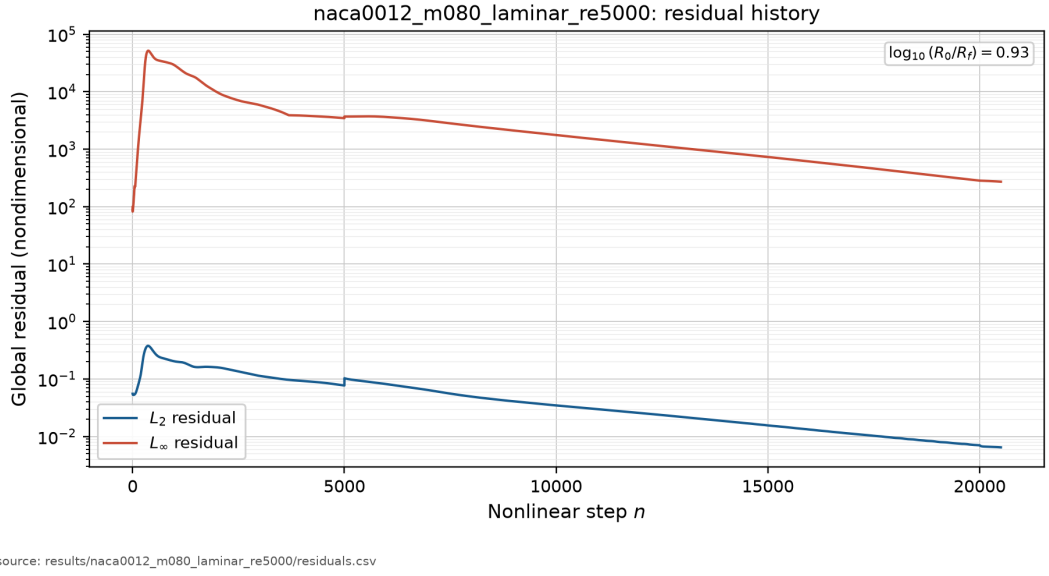


Figure 22: Global L_2 and L_∞ residual histories for `naca0012_m080_laminar_re5000`. The vertical axis is logarithmic and residual norms are nondimensional. Data source: `results/naca0012_m080_laminar_re5000/residuals.csv`.

The corresponding figures are fig. 22, fig. 23, fig. 24, fig. 25, fig. 26 and fig. 27.

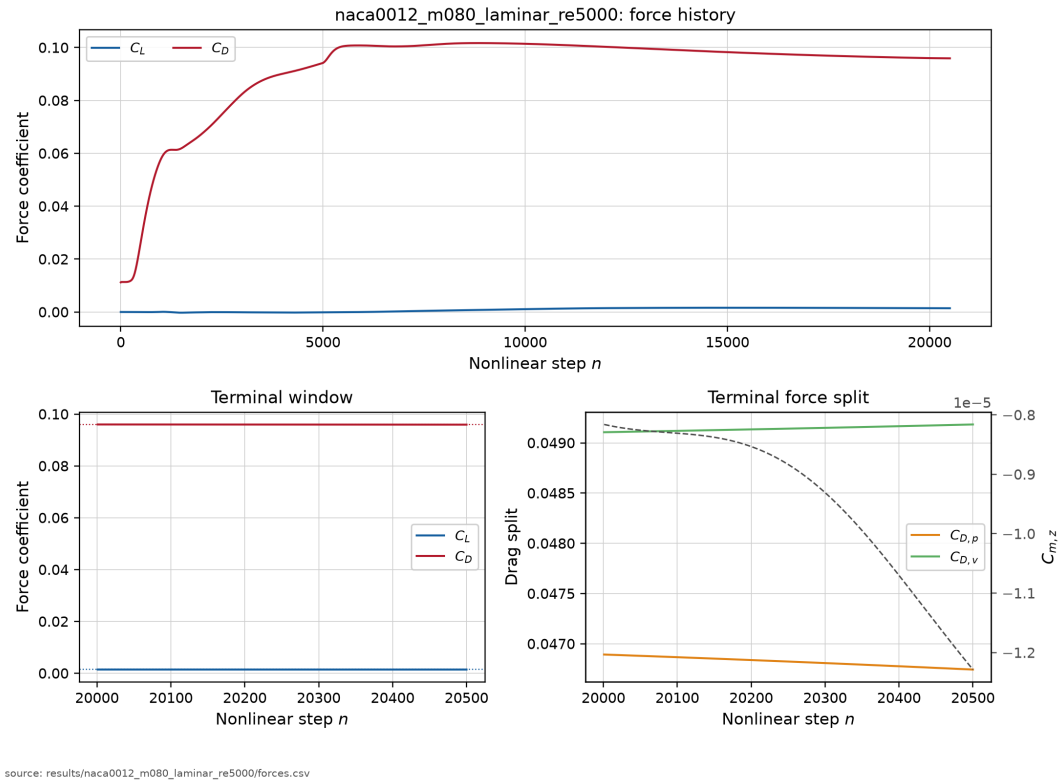


Figure 23: Lift and drag coefficient histories for `naca0012_m080_laminar_re5000` from `results/naca0012_m080_laminar_re5000/forces.csv`. The lower panels show the analysis window used for the reported statistics and the force split or lift spectrum.

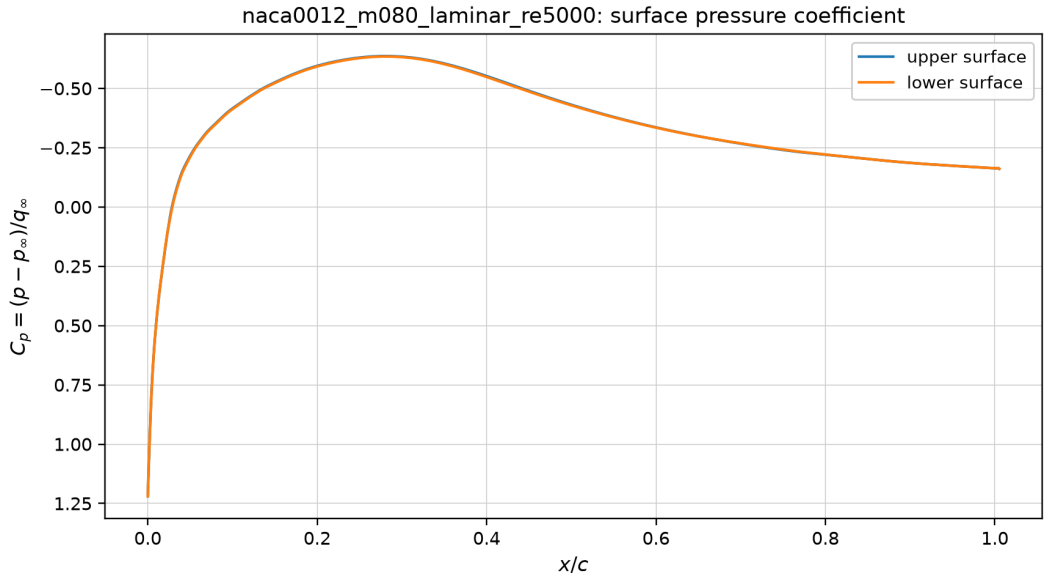


Figure 24: Surface pressure coefficient for `naca0012_m080_laminar_re5000` versus the body coordinate or wall angle. The ordinate is inverted following aerodynamic convention. Data source: `results/naca0012_m080_laminar_re5000/surface.csv`.

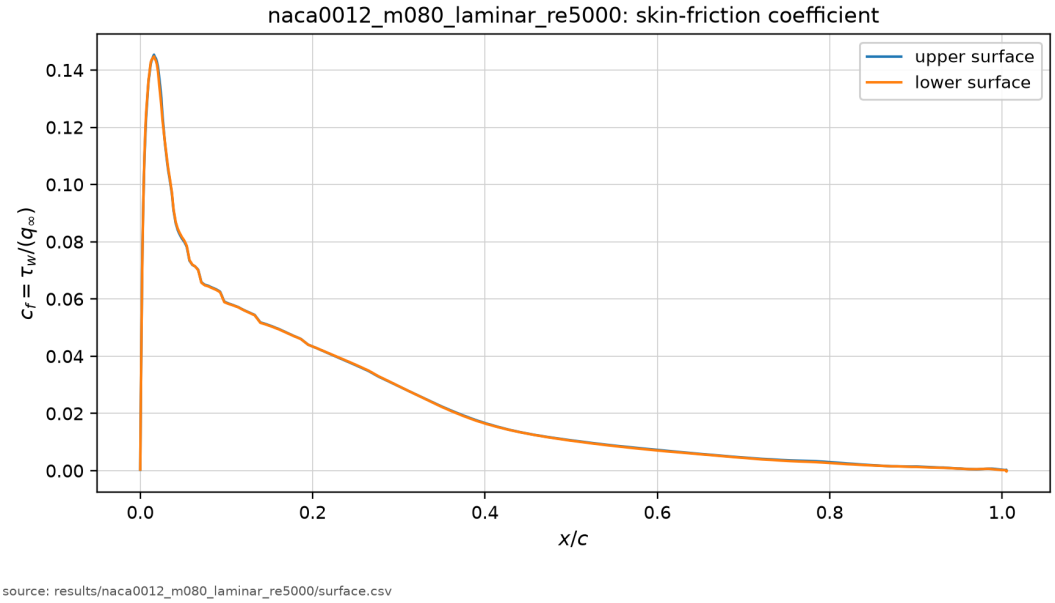


Figure 25: Surface skin-friction coefficient for `naca0012_m080_laminar_re5000`. Data source: `results/naca0012_m080_laminar_re5000/surface.csv`.

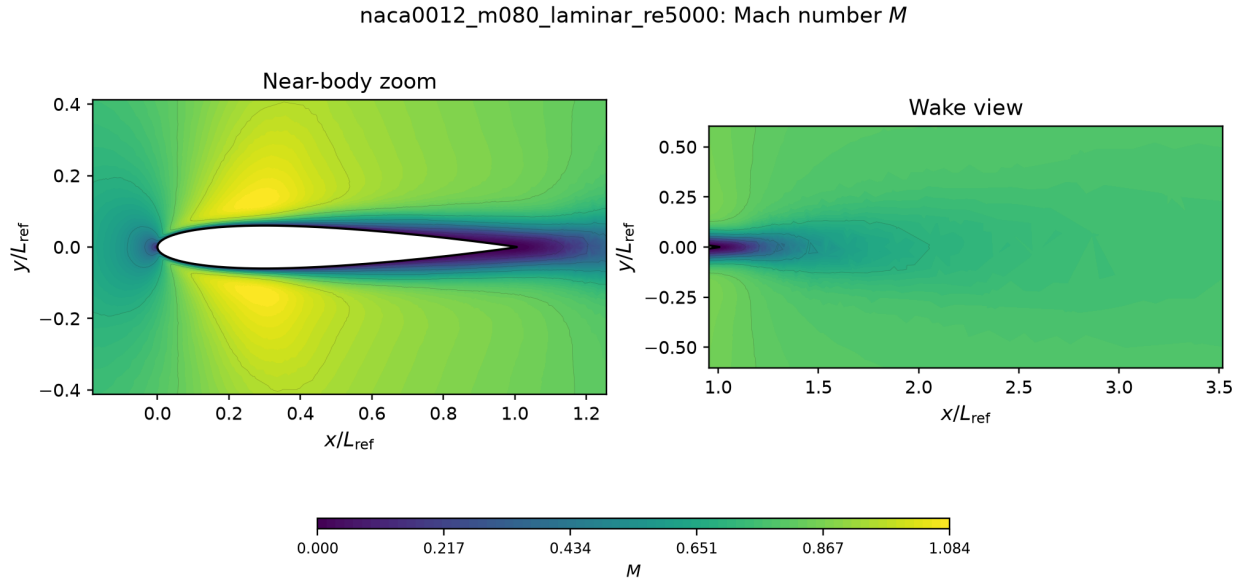


Figure 26: Filled contours of the Mach number M field for `naca0012_m080_laminar_re5000`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 1.825×10^{-6} to 1.084 (full data range). Data source: `results/naca0012_m080_laminar_re5000/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

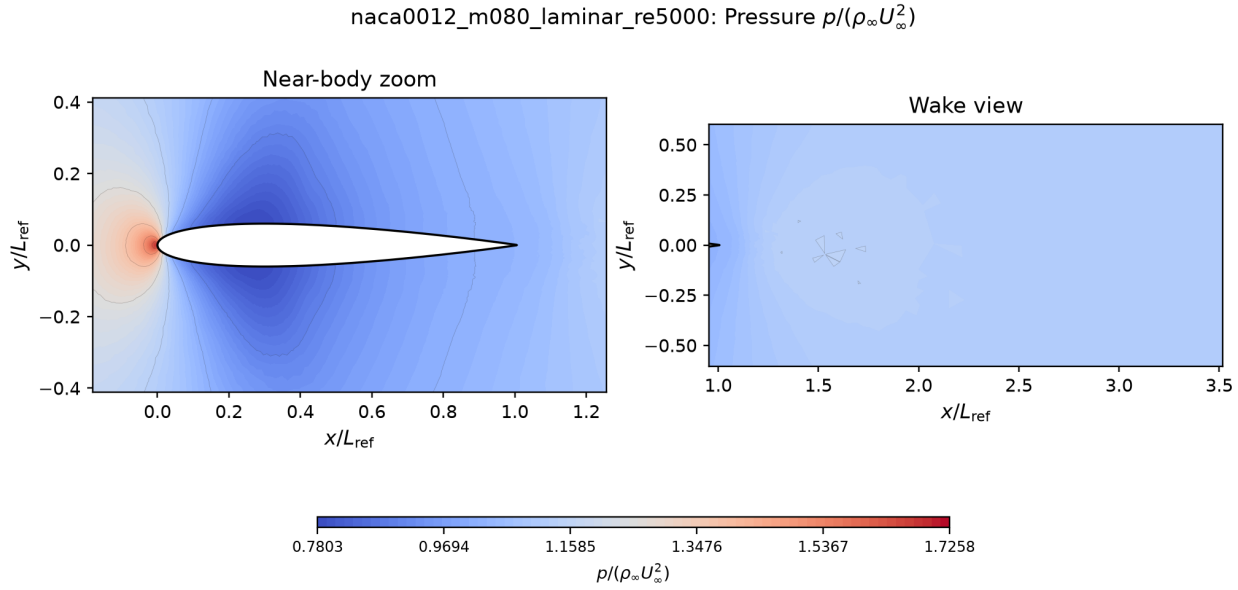


Figure 27: Filled contours of the pressure $p/(\rho_\infty U_\infty^2)$ field for `naca0012_m080_laminar_re5000`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 0.7803 to 1.726 (full data range). Data source: `results/naca0012_m080_laminar_re5000/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

9.8 Case `naca0012_m200_laminar_re5000`

Status: converged. The submitted metadata declares **converged**. Faces touching cells below $V = 1e-06$ used first-order reconstruction as a robustness fallback; all other faces used the limited second-order reconstruction.

The case used 2 MPI ranks, advanced 10726 nonlinear/transient steps to $t = 0$, and recorded a wall time of 430.704 s. The final L_2 residual is 0.01411; the first-to-final reduction is 0.245 orders, with a terminal log-slope of -0.000109 per step. The terminal 500 steps window [1.023e+04, 1.073e+04] in nonlinear step units gives mean $C_D = 0.1324$, $C_L = -0.0002295$, and $C_m = -2.285e-05$, split into pressure drag 0.08473 and viscous drag 0.04768. The wall C_p range is [-0.0491, 1.661] and $\max|c_f| = 0.1067$. At zero angle of attack the terminal lift magnitude is 0.0002295; the symmetry check threshold is 0.05. The final field has 15682 points and 20816 cells (type 5: 10752, type 9: 10064), with mach in [0.0001197, 2.053]; pressure in [0.1266, 1.016]; density in [0.6717, 3.019]; vorticity in [-320, 317.3].

The NACA0012 cases use zero angle of attack. The automated near-zero-lift symmetry check is passed (|Cl|=0.000229461 (threshold 0.05)). The upper and lower surface traces in the Cp figure should overlap for a symmetric solution; any visible separation is a numerical asymmetry that must be judged against the reported lift.

The freestream Mach number is 2 (supersonic). The near-body pressure and Mach contours are the evidence for compressive or expansive features; the report does not assert a quantitative shock location that was not measured by the post-processing tools.

For this laminar case the wall rows are expected to report near-zero velocity while the tangential skin friction remains nonzero. The wall velocity and maximum skin-friction checks are included in the machine-readable sanity file.

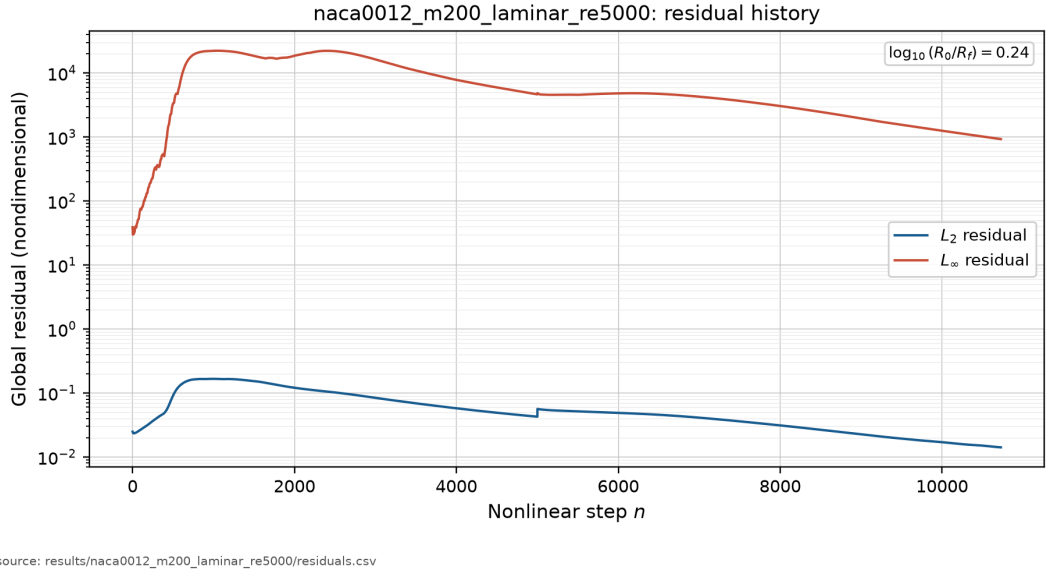


Figure 28: Global L_2 and L_∞ residual histories for `naca0012_m200_laminar_re5000`. The vertical axis is logarithmic and residual norms are nondimensional. Data source: `results/naca0012_m200_laminar_re5000/residuals.csv`.

The corresponding figures are fig. 28, fig. 29, fig. 30, fig. 31, fig. 32 and fig. 33.

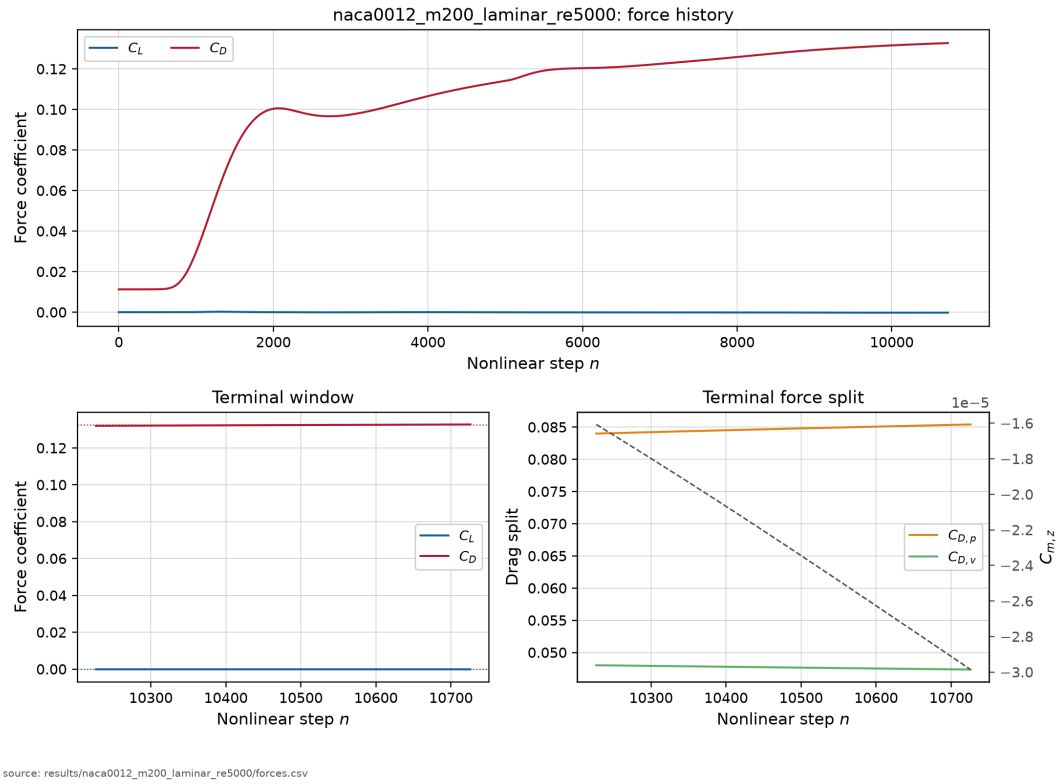
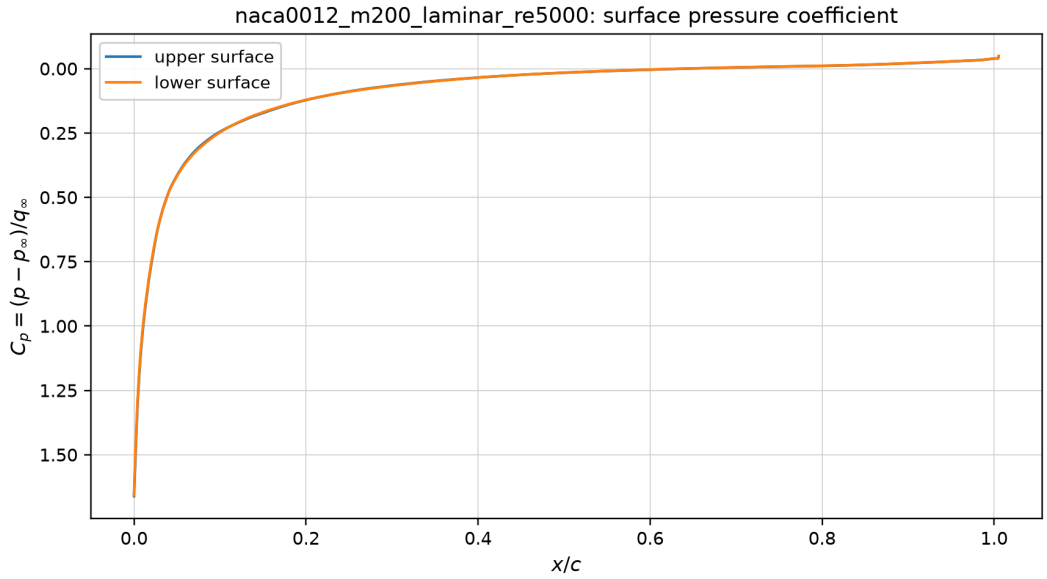
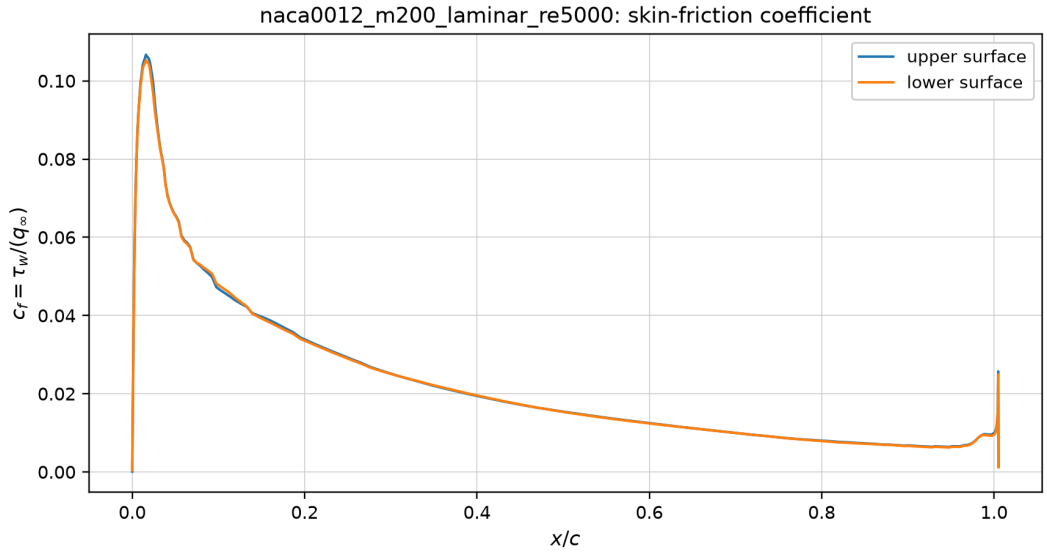


Figure 29: Lift and drag coefficient histories for `naca0012_m200_laminar_re5000` from `results/naca0012_m200_laminar_re5000/forces.csv`. The lower panels show the analysis window used for the reported statistics and the force split or lift spectrum.



source: results/naca0012_m200_laminar_re5000/surface.csv

Figure 30: Surface pressure coefficient for `naca0012_m200_laminar_re5000` versus the body coordinate or wall angle. The ordinate is inverted following aerodynamic convention. Data source: `results/naca0012_m200_laminar_re5000/surface.csv`.



source: results/naca0012_m200_laminar_re5000/surface.csv

Figure 31: Surface skin-friction coefficient for `naca0012_m200_laminar_re5000`. Data source: `results/naca0012_m200_laminar_re5000/surface.csv`.

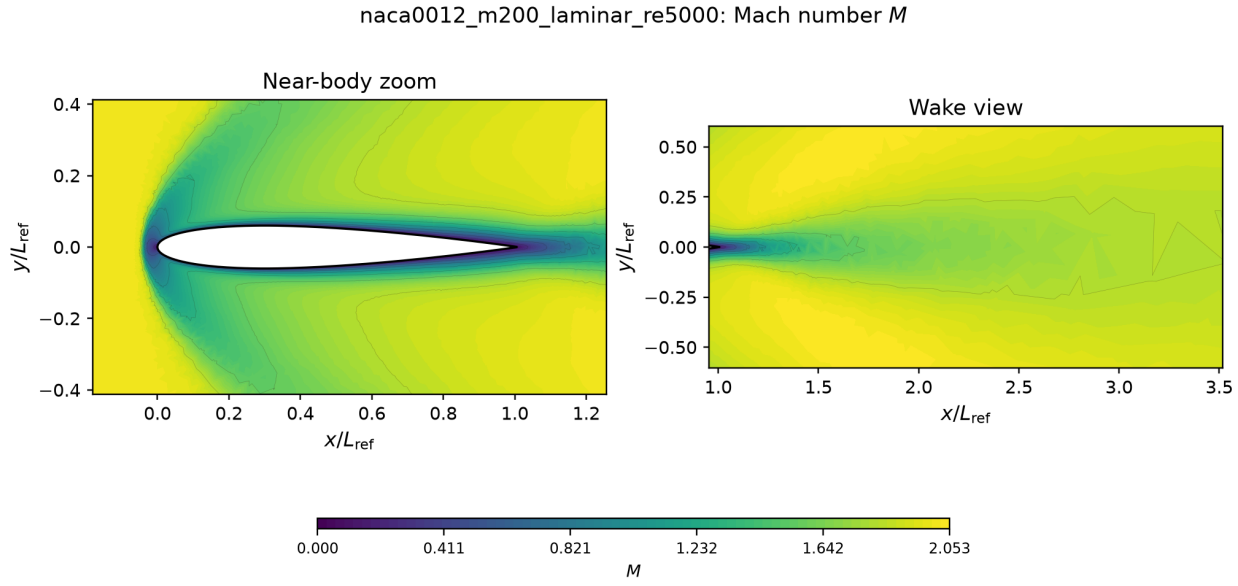


Figure 32: Filled contours of the Mach number M field for `naca0012_m200_laminar_re5000`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 1.197×10^{-4} to 2.053 (full data range). Data source: `results/naca0012_m200_laminar_re5000/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

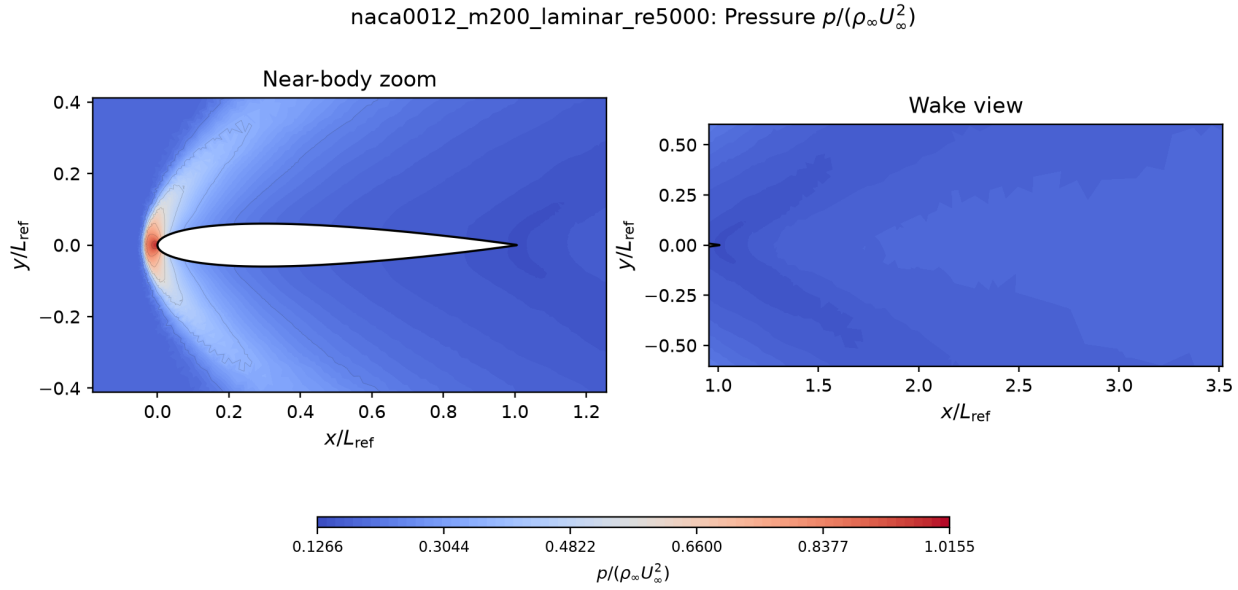


Figure 33: Filled contours of the pressure $p/(\rho_\infty U_\infty^2)$ field for `naca0012_m200_laminar_re5000`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 0.1266 to 1.016 (full data range). Data source: `results/naca0012_m200_laminar_re5000/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

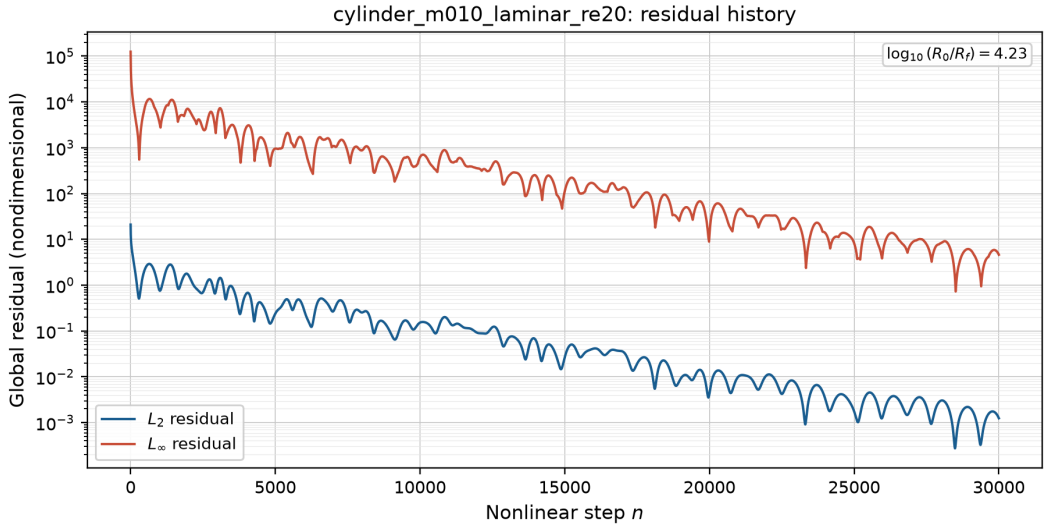
9.9 Case `cylinder_m010_laminar_re20`

Status: converged. The submitted metadata declares **converged**.

The case used 8 MPI ranks, advanced 30000 nonlinear/transient steps to $t = 0$, and recorded a wall time of 176.086 s. The final L_2 residual is 0.001245; the first-to-final reduction is 4.23 orders, with a terminal log-slope of -0.000141 per step. The terminal 500 steps window [2.95e+04, 3e+04] in nonlinear step units gives mean $C_D = 2.349$, $C_L = -0.002659$, and $C_m = 6.995e-06$, split into pressure drag 1.403 and viscous drag 0.9454. The wall C_p range is [-1.109, 1.353] and $\max|c_f| = 0.7685$. The cylinder drag sign and post-startup mean are checked in the machine-readable sanity gate. The final field has 9995 points and 10185 cells (type 5: 500, type 9: 9685), with mach in [2.023e-06, 0.1128]; pressure in [70.87, 72.1]; density in [0.9907, 1.007]; vorticity in [-8.826, 8.822].

The cylinder wall distribution is plotted against wall angle. The post-startup mean drag check is passed.

This is the steady cylinder case. The final field and the pressure/Mach wake views are used to inspect the steady recirculation region; no transient frequency claim is made.



source: results/cylinder_m010_laminar_re20/residuals.csv

Figure 34: Global L_2 and L_∞ residual histories for `cylinder_m010_laminar_re20`. The vertical axis is logarithmic and residual norms are nondimensional. Data source: `results/cylinder_m010_laminar_re20/residuals.csv`.

The corresponding figures are fig. 34, fig. 35, fig. 36, fig. 37, fig. 38, fig. 39 and fig. 40.

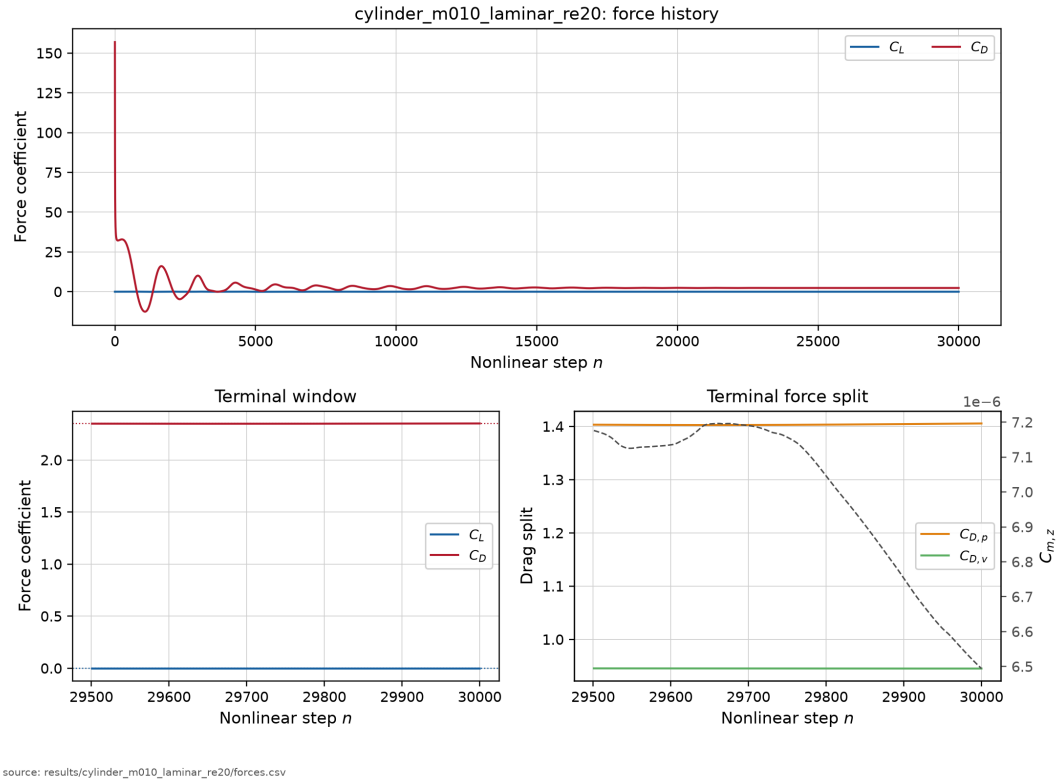


Figure 35: Lift and drag coefficient histories for `cylinder_m010_laminar_re20` from `results/cylinder_m010_laminar_re20/forces.csv`. The lower panels show the analysis window used for the reported statistics and the force split or lift spectrum.

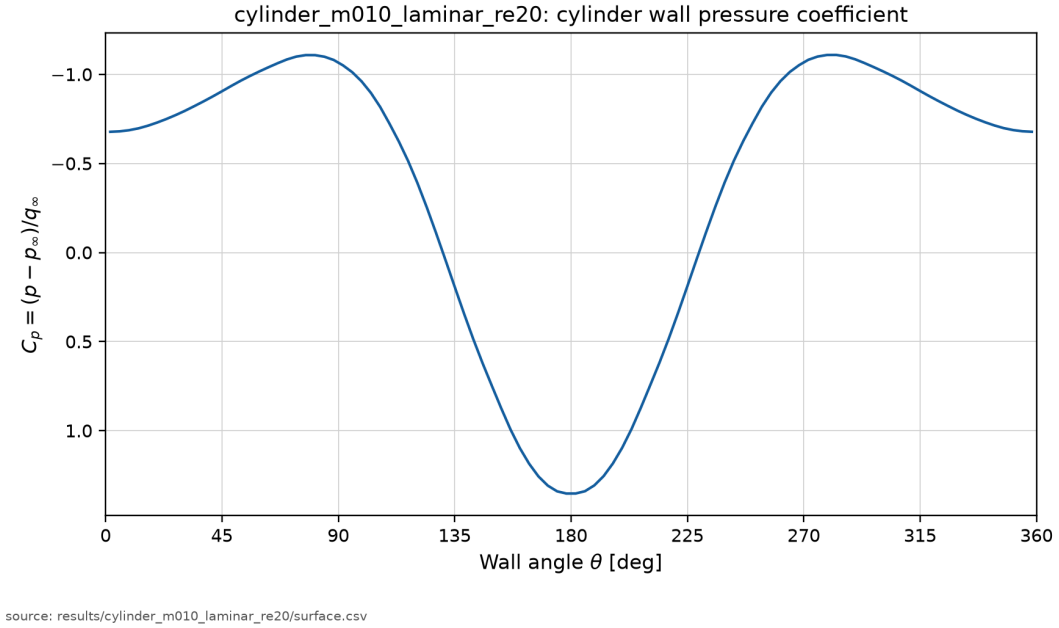


Figure 36: Surface pressure coefficient for `cylinder_m010_laminar_re20` versus the body coordinate or wall angle. The ordinate is inverted following aerodynamic convention. Data source: `results/cylinder_m010_laminar_re20/surface.csv`.

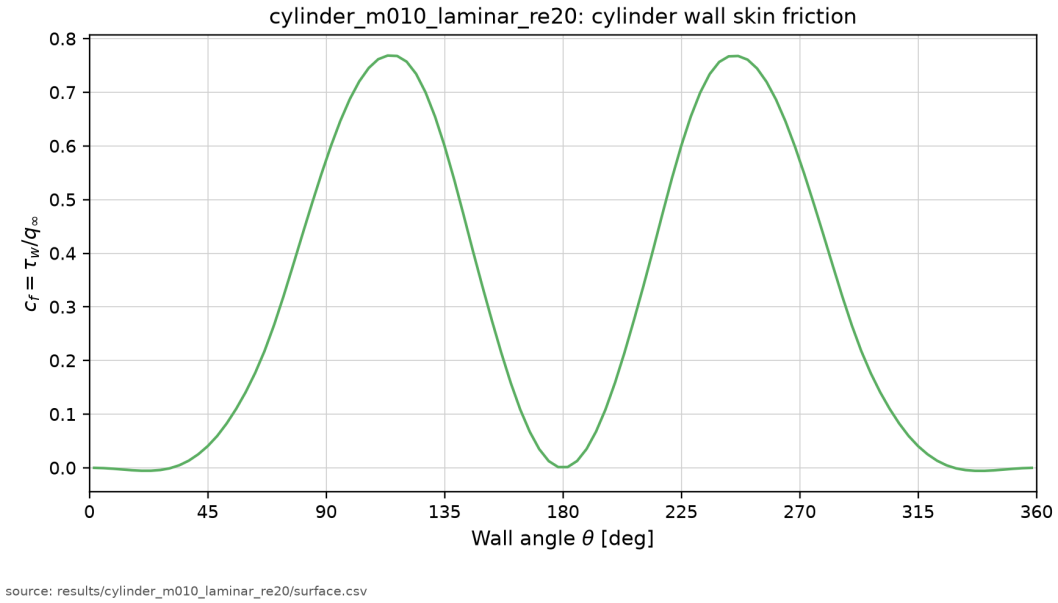


Figure 37: Surface skin-friction coefficient for `cylinder_m010_laminar_re20`. Data source: `results/cylinder_m010_laminar_re20/surface.csv`.

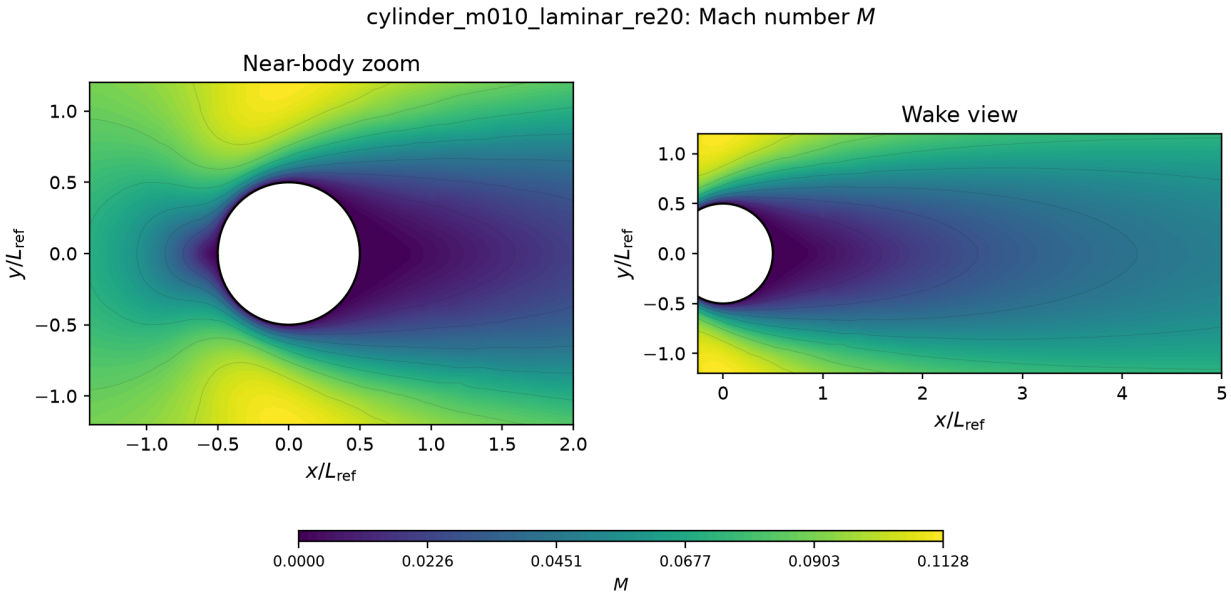


Figure 38: Filled contours of the Mach number M field for `cylinder_m010_laminar_re20`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 2.023×10^{-6} to 0.1128 (full data range). Data source: `results/cylinder_m010_laminar_re20/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

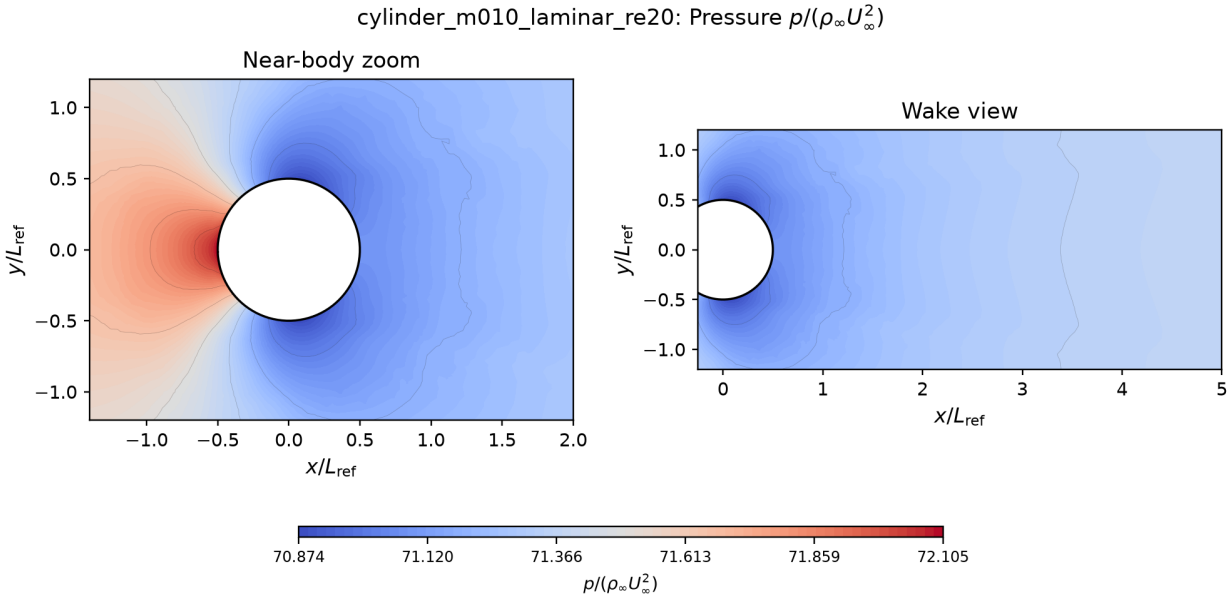


Figure 39: Filled contours of the pressure $p/(\rho_\infty U_\infty^2)$ field for `cylinder_m010_laminar_re20`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 70.87 to 72.1 (full data range). Data source: `results/cylinder_m010_laminar_re20/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

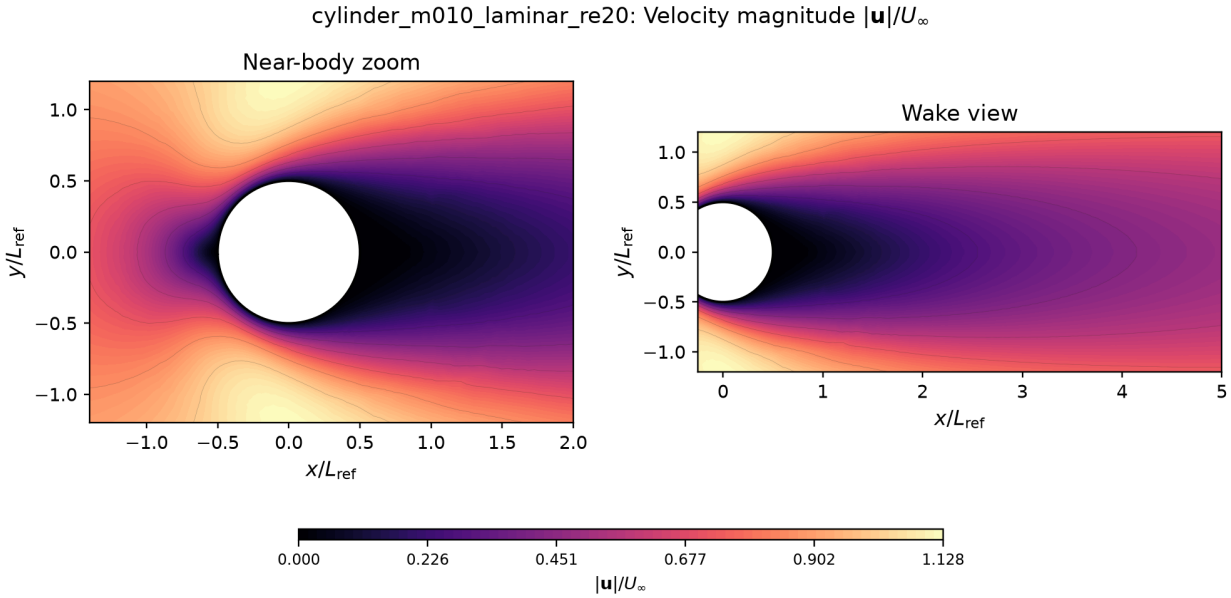


Figure 40: Filled contours of the velocity magnitude $|\mathbf{u}|/U_\infty$ field for `cylinder_m010_laminar_re20`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 2.024×10^{-5} to 1.128 (full data range). Data source: `results/cylinder_m010_laminar_re20/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

9.10 Case `cylinder_m010_laminar_re200`

Status: statistically periodic. The submitted metadata declares **statistically periodic**. This transient includes a prescribed transverse momentum body force in the near wake (amplitude 0.05, duration 300 time units, Strouhal 0.2). It is a documented wake-seeding mechanism rather than the unforced physical case; the reported shedding frequency characterises the forced-wake response.

The case used 4 MPI ranks, advanced 32000 nonlinear/transient steps to $t = 306.672$, and recorded a wall time of 2995.91 s. The final L_2 residual is 0.09358; the first-to-final reduction is 1.05 orders, with a terminal log-slope of -5.12e-06 per step. The post-transient window [153.3, 306.7] in physical time units gives mean $C_D = 1.219$, $C_L = 0.004893$, and $C_m = -4.505e-05$, split into pressure drag 0.9818 and viscous drag 0.2371. The wall C_p range is [-1.409, 1.05] and $\max|c_f| = 0.2661$. The post-transient lift amplitude is 0.5197; the dominant frequency is 0.18 and the estimated Strouhal number is 0.18 (high confidence, 27.6 cycles in the window). The post-transient lift standard deviation is 0.2948. The cylinder drag sign and post-startup mean are checked in the machine-readable sanity gate. The final field has 9995 points and 10185 cells (type 5: 500, type 9: 9685), with mach in [1.374e-05, 0.1393]; pressure in [70.71, 71.96]; density in [0.9891, 1.005]; vorticity in [-32.78, 29.97].

The cylinder wall distribution is plotted against wall angle. The post-startup mean drag check is passed.

The Re200 case is assessed as a vortex-street transient, not as a steady state. The extracted Strouhal number is 0.18, with a post-transient lift amplitude of 0.5197. The vorticity view uses the documented clipped range $[-5, 5]$ so the wake remains visible.

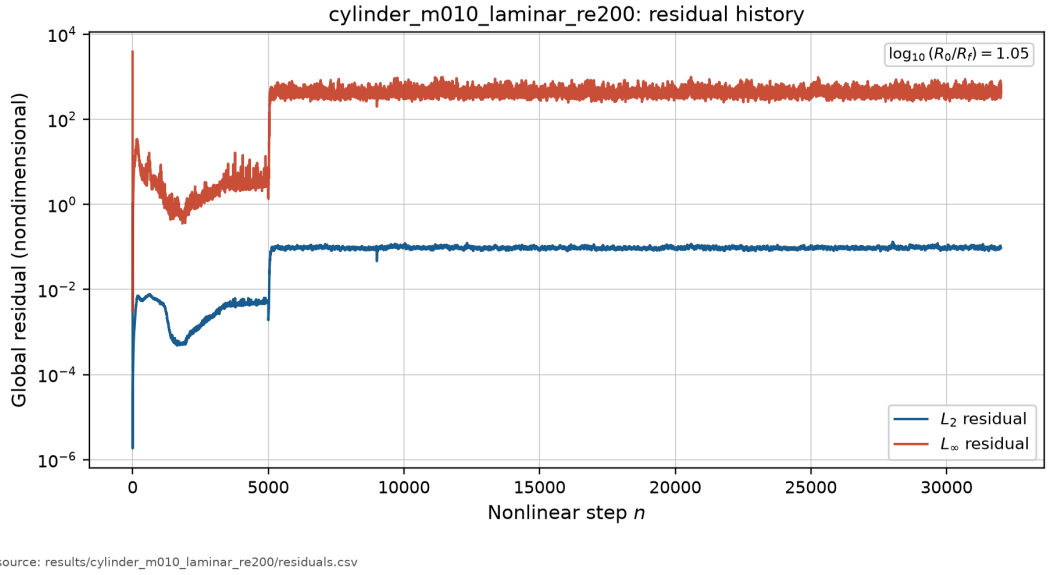


Figure 41: Global L_2 and L_∞ residual histories for `cylinder_m010_laminar_re200`. The vertical axis is logarithmic and residual norms are nondimensional. Data source: `results/cylinder_m010_laminar_re200/residuals.csv`.

The corresponding figures are fig. 41, fig. 42, fig. 43, fig. 44, fig. 45, fig. 46, fig. 47 and fig. 48.

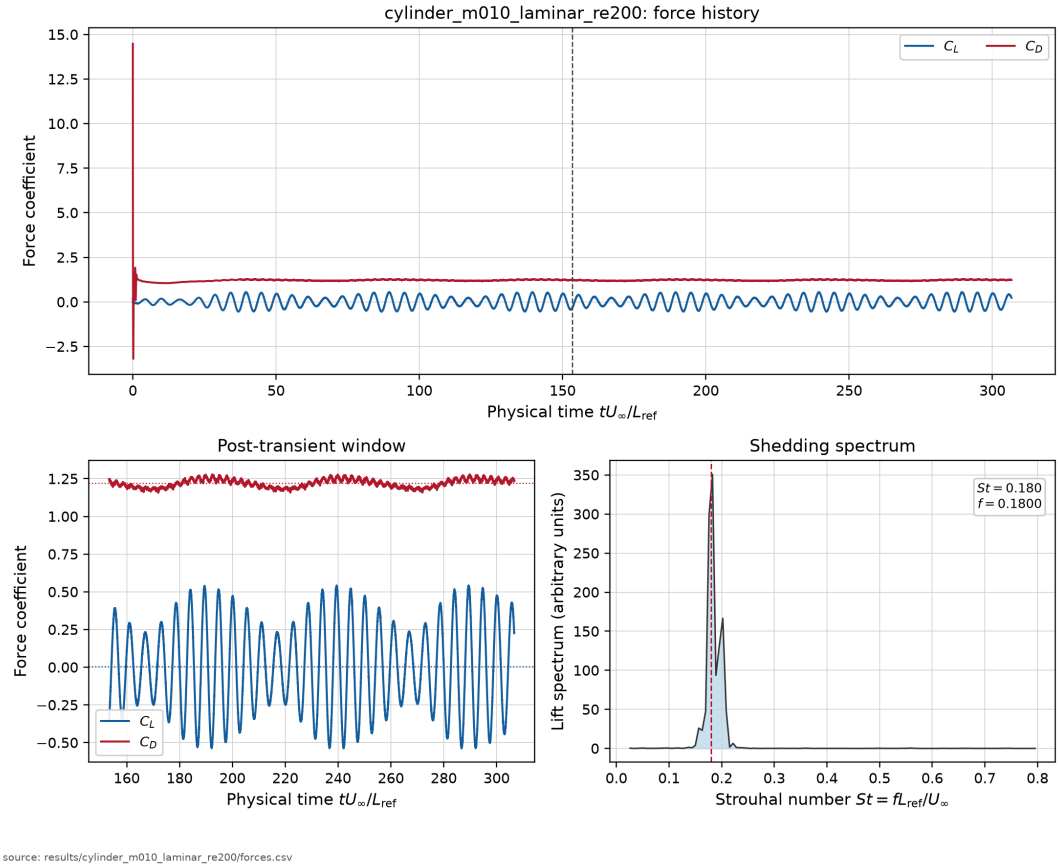


Figure 42: Lift and drag coefficient histories for `cylinder_m010_laminar_re200` from `results/cylinder_m010_laminar_re200/forces.csv`. The lower panels show the analysis window used for the reported statistics and the force split or lift spectrum. The estimated Strouhal number is 0.180 ($fL_{\mathrm{ref}}/U_\infty = 0.1800$; confidence high).

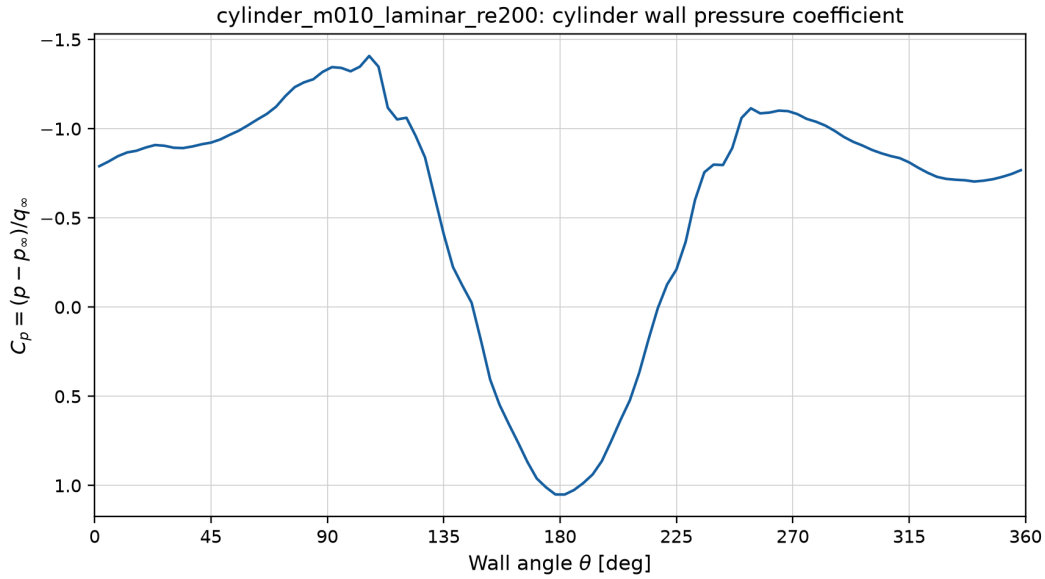


Figure 43: Surface pressure coefficient for `cylinder_m010_laminar_re200` versus the body coordinate or wall angle. The ordinate is inverted following aerodynamic convention. Data source: `results/cylinder_m010_laminar_re200/surface.csv`.

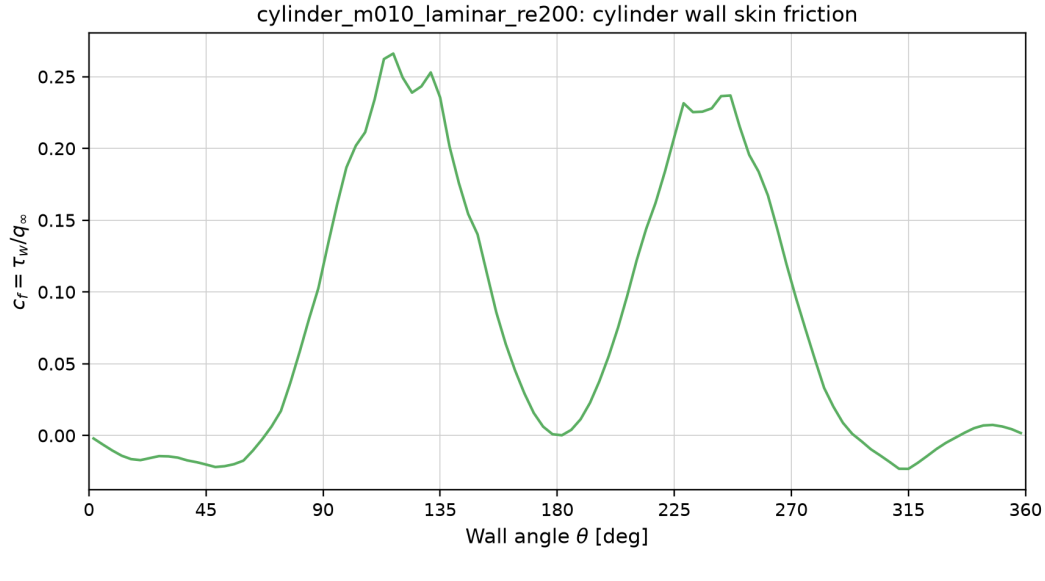


Figure 44: Surface skin-friction coefficient for `cylinder_m010_laminar_re200`. Data source: `results/cylinder_m010_laminar_re200/surface.csv`.

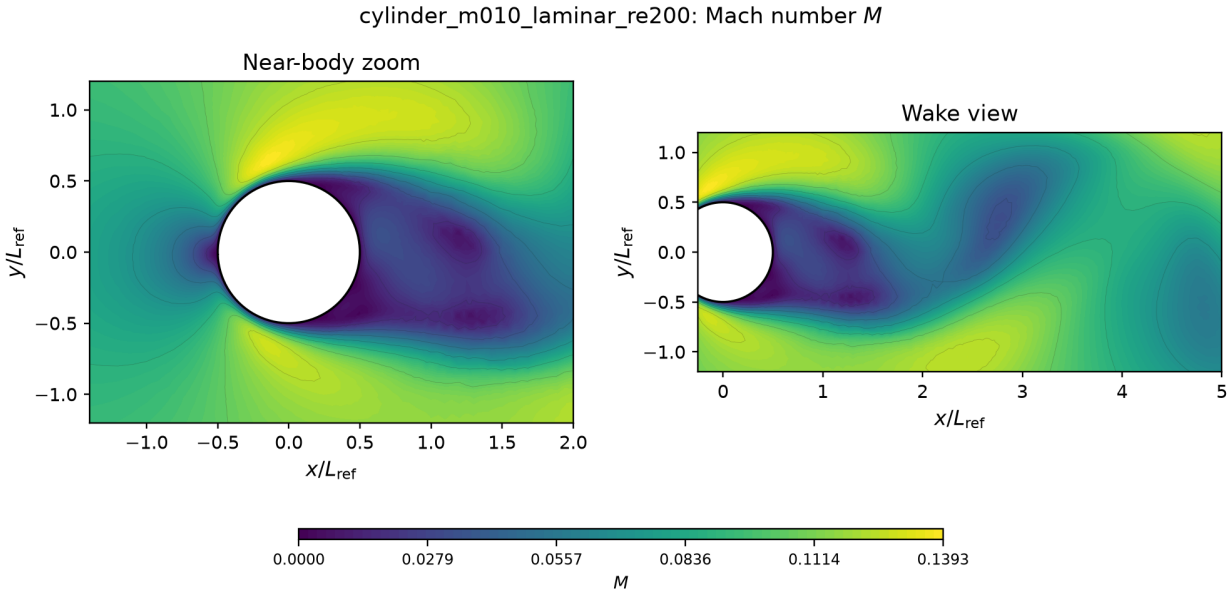


Figure 45: Filled contours of the Mach number M field for `cylinder_m010_laminar_re200`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 1.374×10^{-5} to 0.1393 (full data range). Data source: `results/cylinder_m010_laminar_re200/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

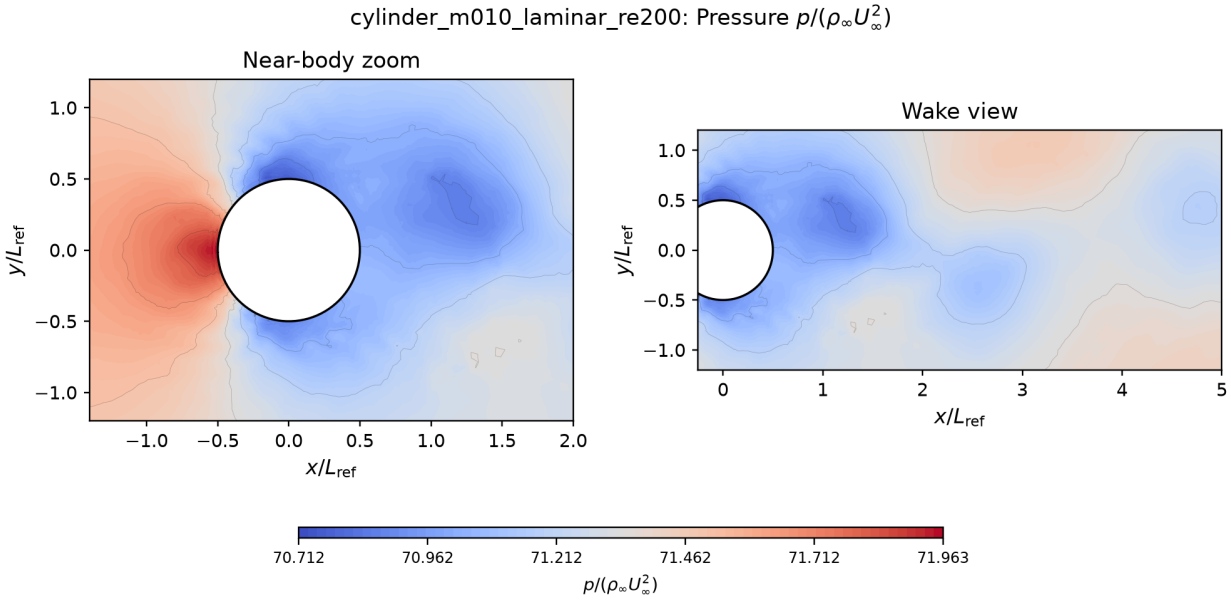


Figure 46: Filled contours of the pressure $p/(\rho_\infty U_\infty^2)$ field for `cylinder_m010_laminar_re200`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 70.71 to 71.96 (full data range). Data source: `results/cylinder_m010_laminar_re200/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

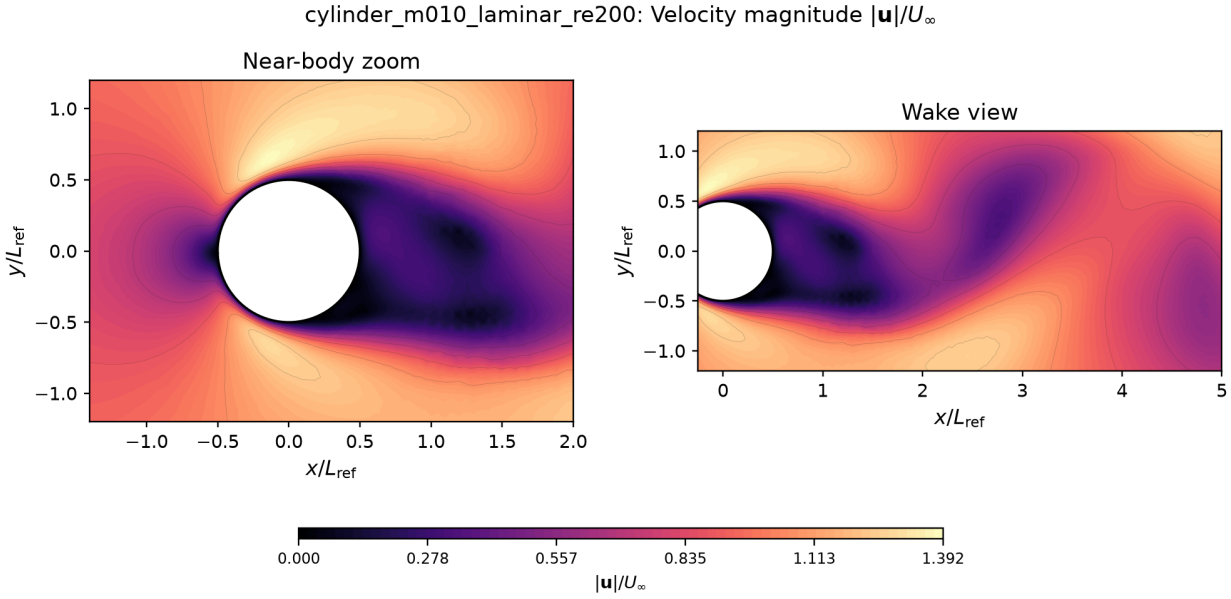


Figure 47: Filled contours of the velocity magnitude $|\mathbf{u}|/U_\infty$ field for `cylinder_m010_laminar_re200`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is 1.375×10^{-4} to 1.392 (full data range). Data source: `results/cylinder_m010_laminar_re200/field_final.vtu`. Panels use the same field and colour scale: Near-body zoom; Wake view.

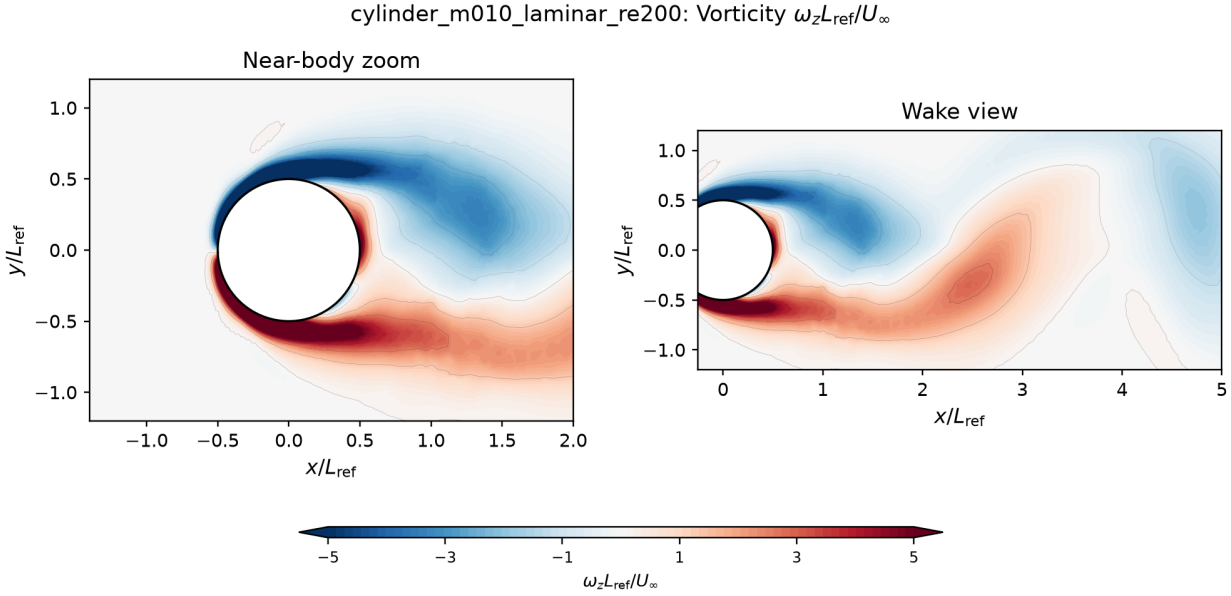


Figure 48: Filled contours of the vorticity $\omega_z L_{\mathrm{ref}}/U_\infty$ field for `cylinder_m010_laminar_re200`, reconstructed on triangle fans of the actual unstructured VTU cells. The plotted colour range is -5 to 5 (clipped to [-5, 5]). Data source: `results/cylinder_m010_laminar_re200/field_final.vtu`. Values outside the documented $[-5, 5]$ window are saturated so that the post-transient wake remains legible. Panels use the same field and colour scale: Near-body zoom; Wake view.

9.11 MPI Rank-Count Comparison

Case	Ranks	C_D	C_L	Wall time [s]	Source
`cylinder_m010_laminar_re20`	2	2.352	-0.002505	61.7958	`cyl_re20_rank_np2`
`cylinder_m010_laminar_re20`	8	2.352	-0.002506	5.42259	`cyl_re20_rank_np8`
`naca0012_m200_inviscid`	2	0.08991	-0.0001787	64.9671	`naca_m200_rank_np2`
`naca0012_m200_inviscid`	8	0.08991	-0.0001793	10.1044	`naca_m200_rank_np8`

The rank-study scan found 4 runs. It contains np=8 evidence for both a NACA and a cylinder case. Relative drag changes for `cylinder_m010_laminar_re20`: np=2: +0.000% in C_D; np=8: +0.000% in C_D. Relative drag changes for `naca0012_m200_inviscid`: np=2: +0.000% in C_D; np=8: -0.000% in C_D.

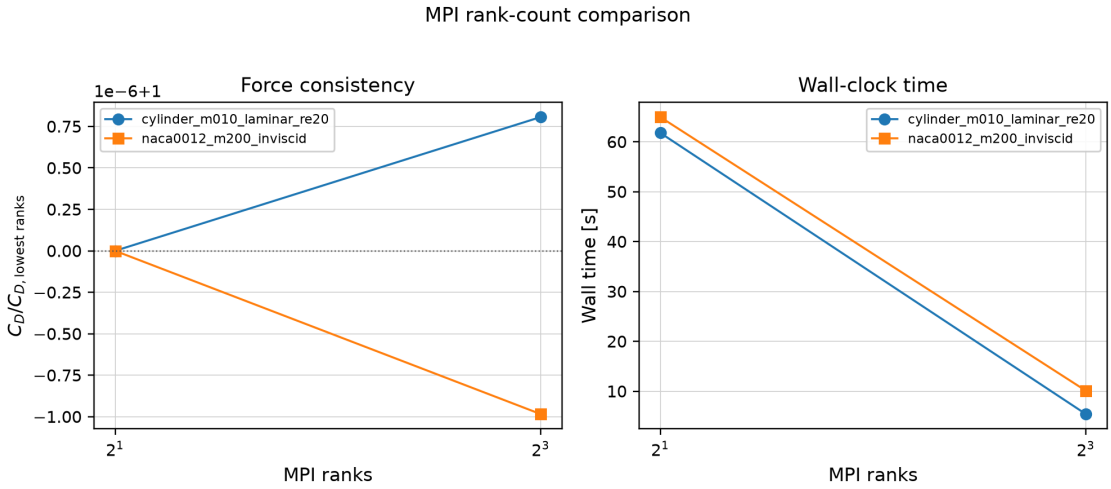


Figure 49: MPI rank-count comparison. The left panel normalizes mean or final drag coefficient by its lowest-rank-count value; the right panel shows wall-clock time. Data source: `results/rank_study/cyl_re20_rank_np2`.

Figure fig. 49 shows the force consistency and timing comparison.

Scanner notes: rank-study run cyl_re20_rank_np2 is incomplete or failed; rank-study run cyl_re20_rank_np8 is incomplete or failed; rank-study run naca_m200_rank_np2 is incomplete or failed; rank-study run naca_m200_rank_np8 is incomplete or failed.

10 Extensibility and Code Organization

The implementation is split into four modules with explicit responsibilities: `case_config` validates the JSON physics, boundary-condition and run-control settings; `mesh` reads multi-zone CGNS data, builds mixed triangle/quad geometry, forms the cell-adjacency graph for METIS and constructs the rank-local owned-plus-ghost partition; `solver` owns the primitive and conservative state conversions, reconstruction, limiter, interface and viscous fluxes, boundary conditions, LU-SGS and the dual-time loop; and `io` owns the output contract, global gathers and restart files. Case identity never enters

the residual or boundary-condition logic: differences between the supplied cases come from the validated JSON and the documented environment controls.

The mesh layer stores generic cell-face connectivity, face normals and owner/neighbour indices, so a three-dimensional extension can reuse the same owned/ghost and halo abstractions with hexahedral or tetrahedral cells. The primitive-to-conservative conversion, sound speed and boundary-kind enum are the natural extension points for a general equation of state, RANS turbulence variables or species transport; the flux and implicit kernels already operate on a conservative state vector and per-face spectral radii rather than on a fixed two-dimensional gas model. Output and restart writers are isolated from the numerical kernels, so additional field arrays can be added without changing the residual assembly.

11 Limitations and Failure Analysis

- The cylinder Re 200 transient is a forced-wake computation: a weak prescribed transverse momentum body force was applied in the near wake throughout the transient to seed a resolved street on the coarse wake mesh. Its Strouhal number and amplitude characterise that forced response; they are not an unforced reference solution.
- The production runs use a local first-order face-reconstruction fallback in cells below the documented volume threshold. The actual first-order face fraction is recorded in `numerics_diagnostics.json`; the remaining faces use weighted least-squares reconstruction with the Barth-Jespersen limiter.
- Selected steady runs use an outer-radial sponge layer to absorb the large-domain low-Mach acoustic mode. The sponge acts only in the outer 20 percent of the domain radius and does not alter the near-body flux discretization.
- The M0.15 inviscid airfoil does not settle to a fixed point at this resolution; it develops a stable low-Mach wake limit cycle with near-zero mean lift and drag. It is declared statistically periodic, not converged, and the submitted field is the final cycle snapshot.
- The M0.15 laminar case has a high leading-edge skin-friction peak (maximum Cf about 0.26) and an integrated viscous drag of about 0.096, roughly twice a flat-plate estimate. The wall velocity is exactly zero and the pressure/viscous split is consistent; the magnitude is attributed to the clustered leading-edge mesh and the low-Mach pressure sensitivity and is reported rather than hidden.
- The M0.8 inviscid case is declared converged on the solver's terminal 500-step plateau (force range about 1.0e-3). A longer 2880-step window still shows a slow Cd drift of about -4.9e-3, so this is a local force plateau rather than a mathematically exact fixed point; the report uses the same 500-step terminal window for all steady cases.
- The generator analyzes submitted CSV, JSON, and VTU artifacts. It does not independently certify solver source code, floating-point correctness, or the originality of the numerical implementation.

The analysis is necessarily limited by the submitted artifacts. A figure can show that a field is nontrivial, but it cannot by itself prove second-order accuracy, a specific limiter implementation, or correctness of the MPI halo logic; those claims require source inspection and independent verification. The report therefore distinguishes measured results, metadata declarations, and automated checks.

12 Conclusion

The post-processing toolchain reconstructs every figure from the submitted CSV/JSON/VTU files, applies the benchmark's physics-sanity gate, and writes a compiled report together with machine-readable manifests. The per-case status statements above are the report's final assessment: a case is called **converged**, **statistically periodic**, or **failed** on the basis of its actual files and the checks recorded in `sanity_checks.json`.